23rd Annual Meeting of the Complex Trait Community

Workshop: Tools For Variant Visualization and Analysis at the Rat Genome Database (RGD) and the Alliance of Genome Resources (the Alliance)

Jennifer R. Smith

Rat Genome Database

https://rgd.mcw.edu

Alliance of Genome Resources

https://www.alliancegenome.org/

https://rgd.mcw.edu

MEDICAL COLLEGE OF WISCONSIN

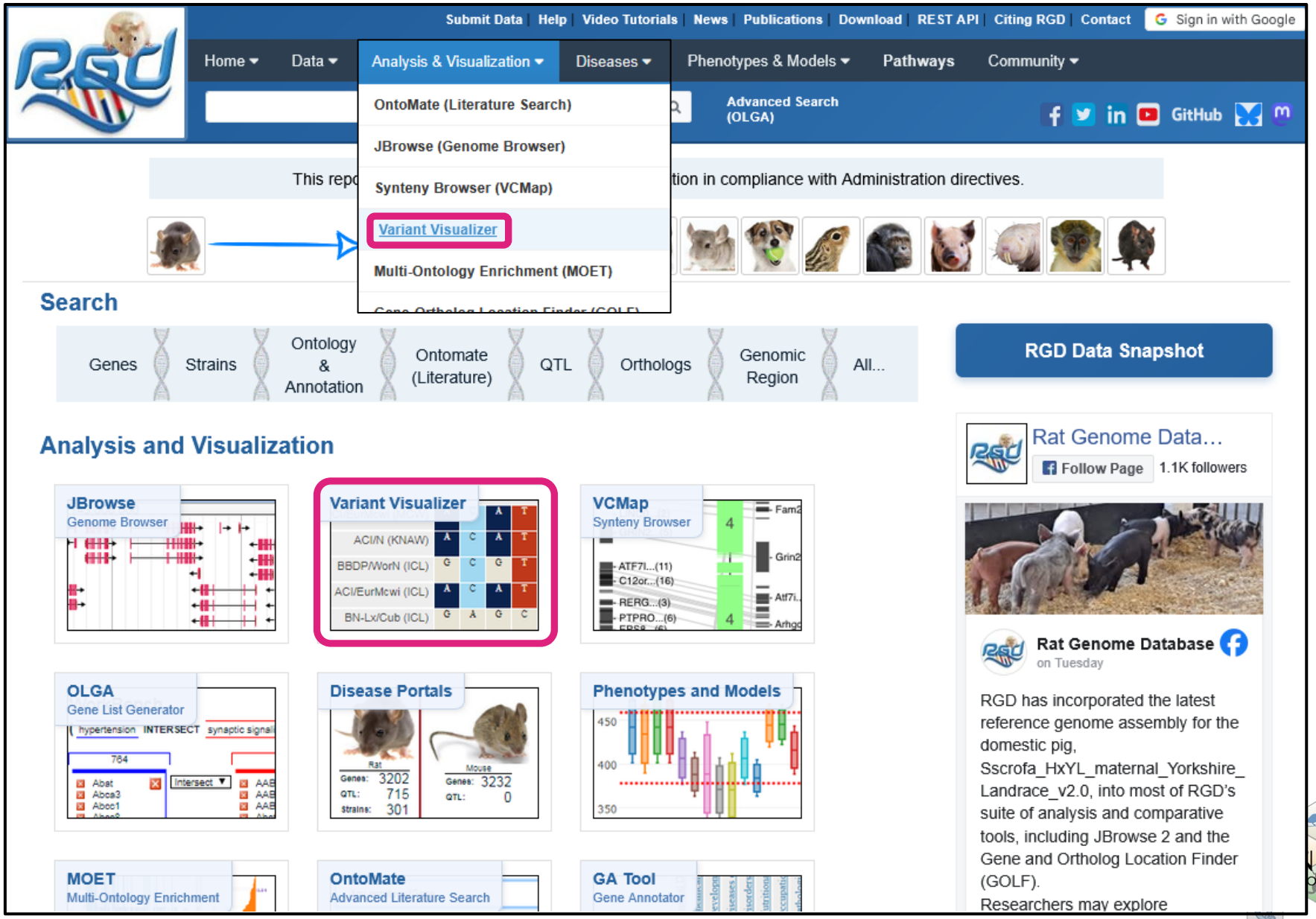
physiology

GBC GLOBAL CORE BIODATA RESOURCE

ALLIANCE of GENOME RESOURCES

RGD: Accessing Variant Visualizer

Access Variant Visualizer from the dropdown "Analysis & Visualization" menu in the main menu bar at the top of any RGD page or use the "Variant Visualizer" button on the RGD home page.

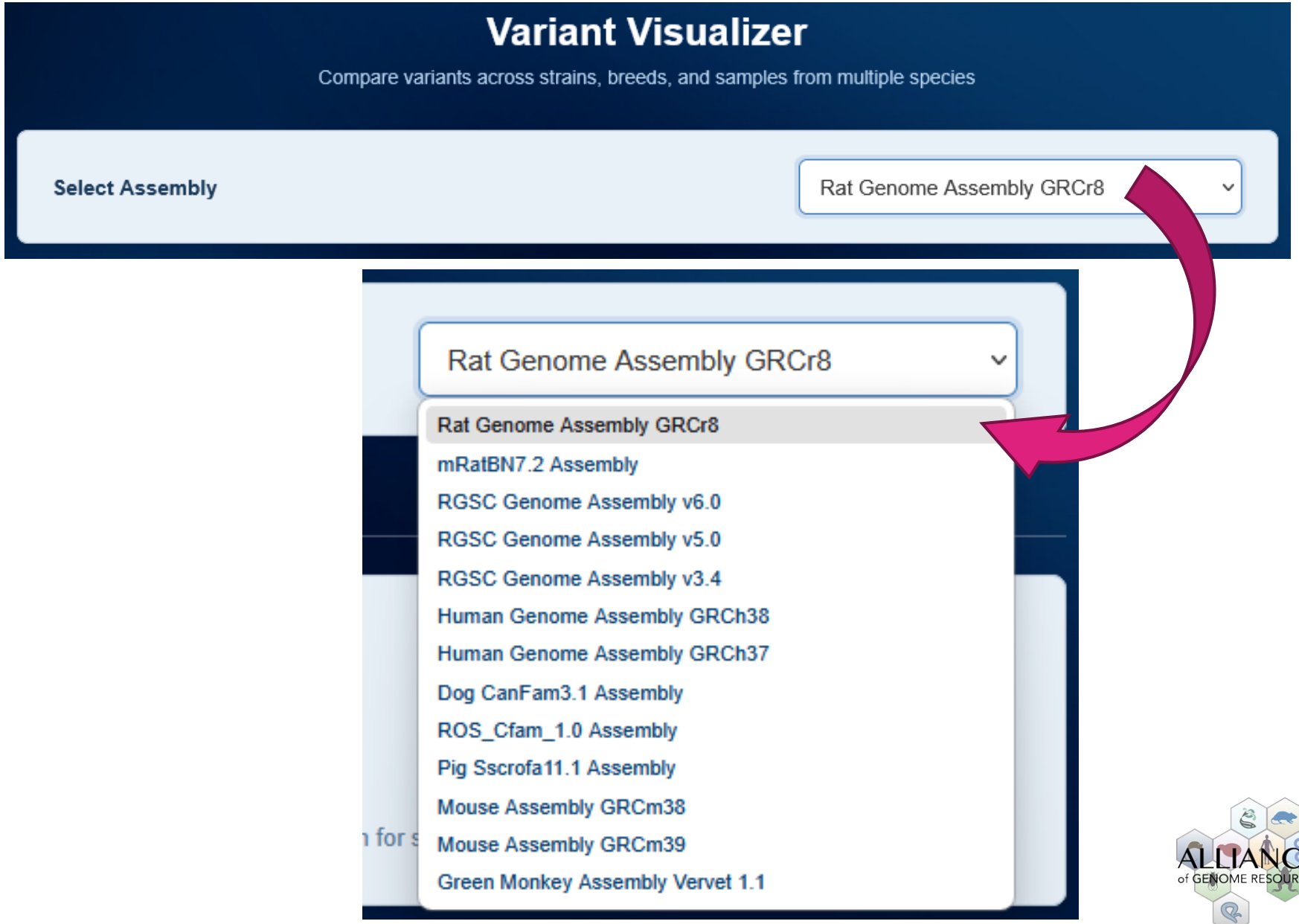


https://rgd.mcw.edu

RGD: The Variant Visualizer Landing Page

Start a search in Variant Visualizer by selecting a species and an assembly.

For this example we will use GRCr8.



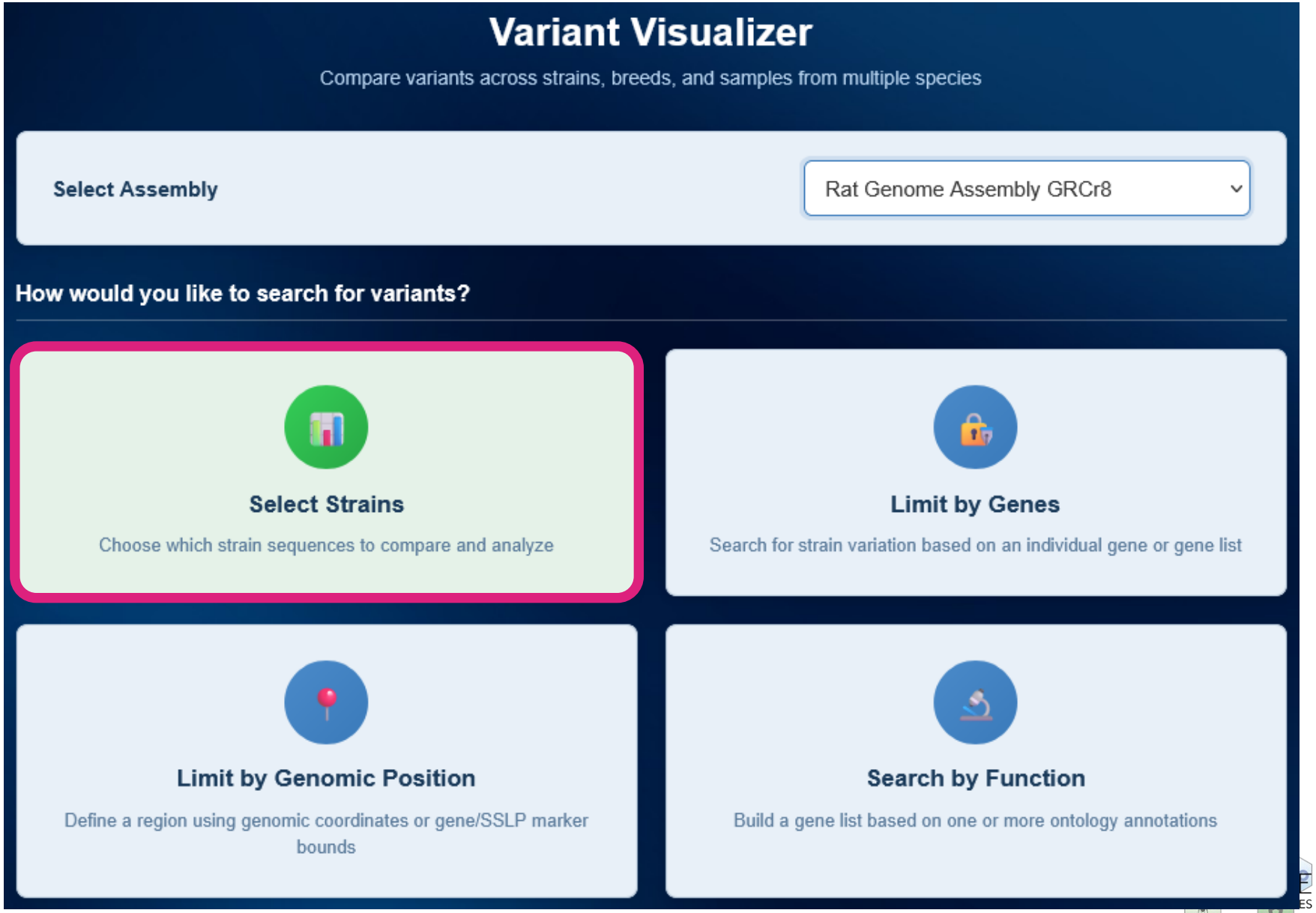
Variant Visualizer

https://rgd.mcw.edu

RGD: The Variant Visualizer Landing Page

Once the species and assembly are selected, you have several options for searching for variants, any of which can be selected first.

For this example, we will begin by selecting strains. The other options will be available again later.



Variant Visualizer

https://rgd.mcw.edu

RGD: The Variant Visualizer Strain/Sample Selection

- The strain/sample selection page has been completely redesigned.
- Sequence groups are listed at the top.
- Below that the samples are grouped by strain.
- These are closed by default.

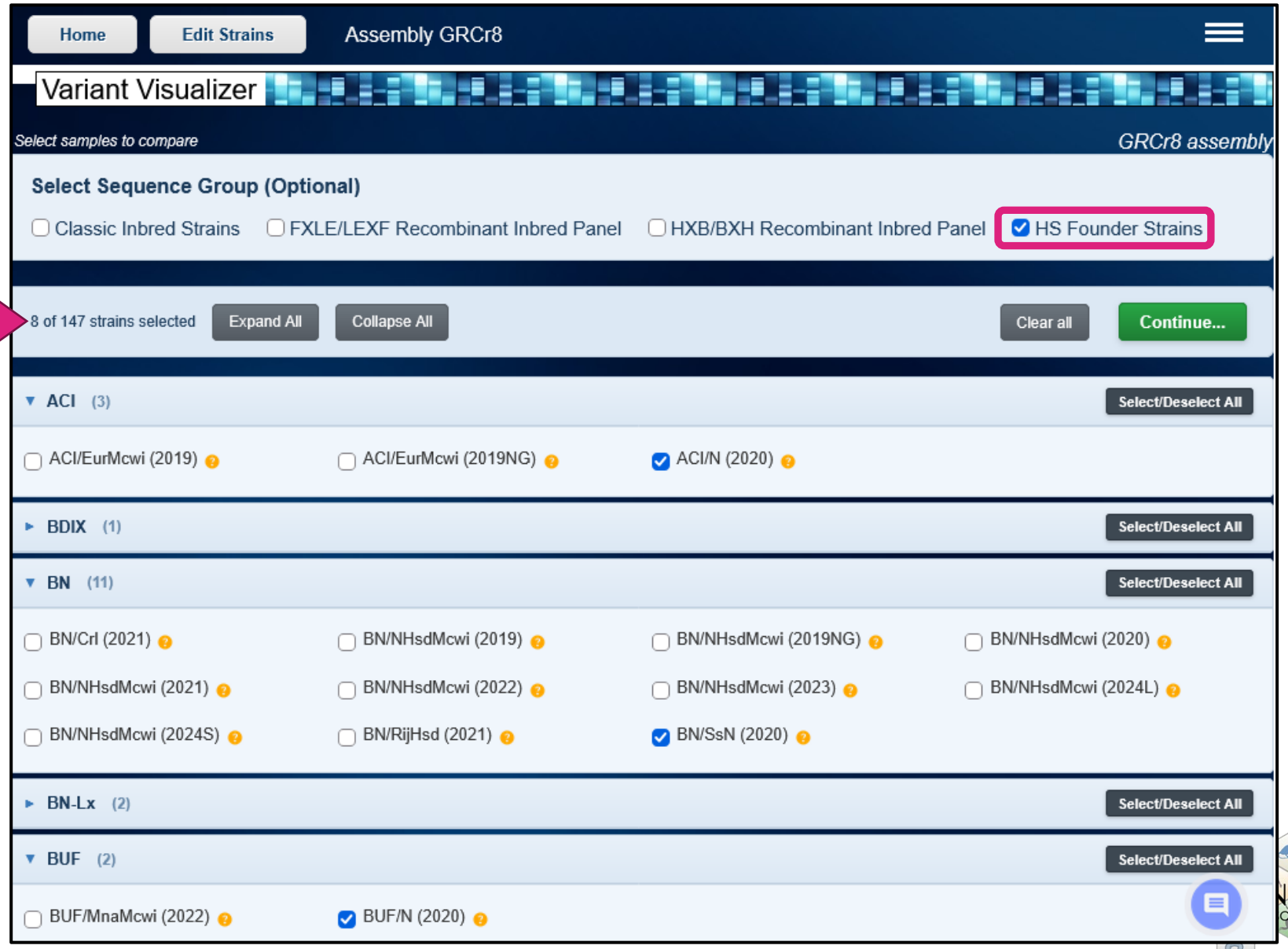


https://rgd.mcw.edu

RGD: The Variant Visualizer Strain/Sample Selection

Click a collapsed row to open it or select a Sequence Group in the top list to open the rows containing samples in that group.

Selecting the HS Founder Strains group automatically selects 8 of the 147 available samples and opens the corresponding rows

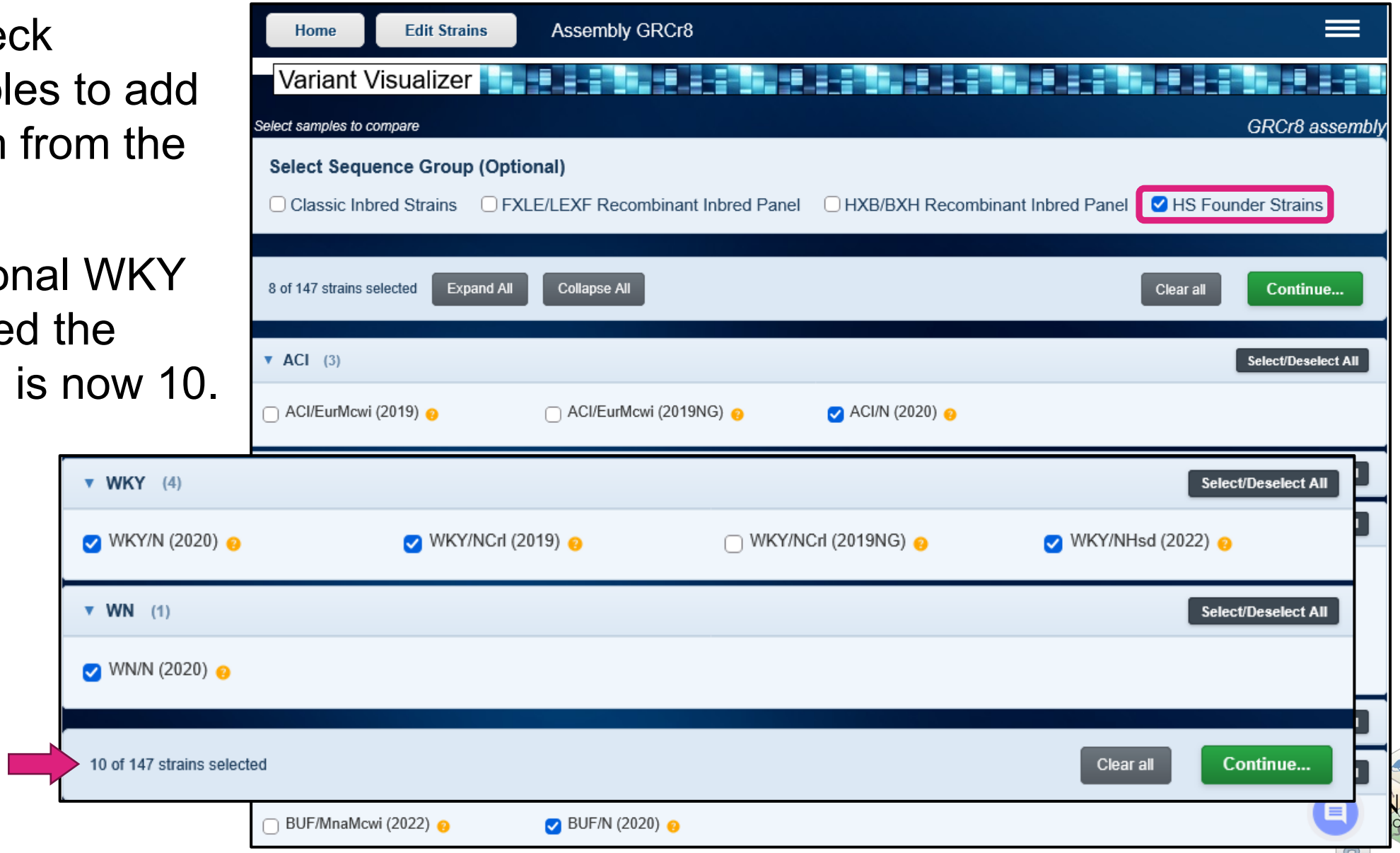


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RGD: The Variant Visualizer Strain/Sample Selection

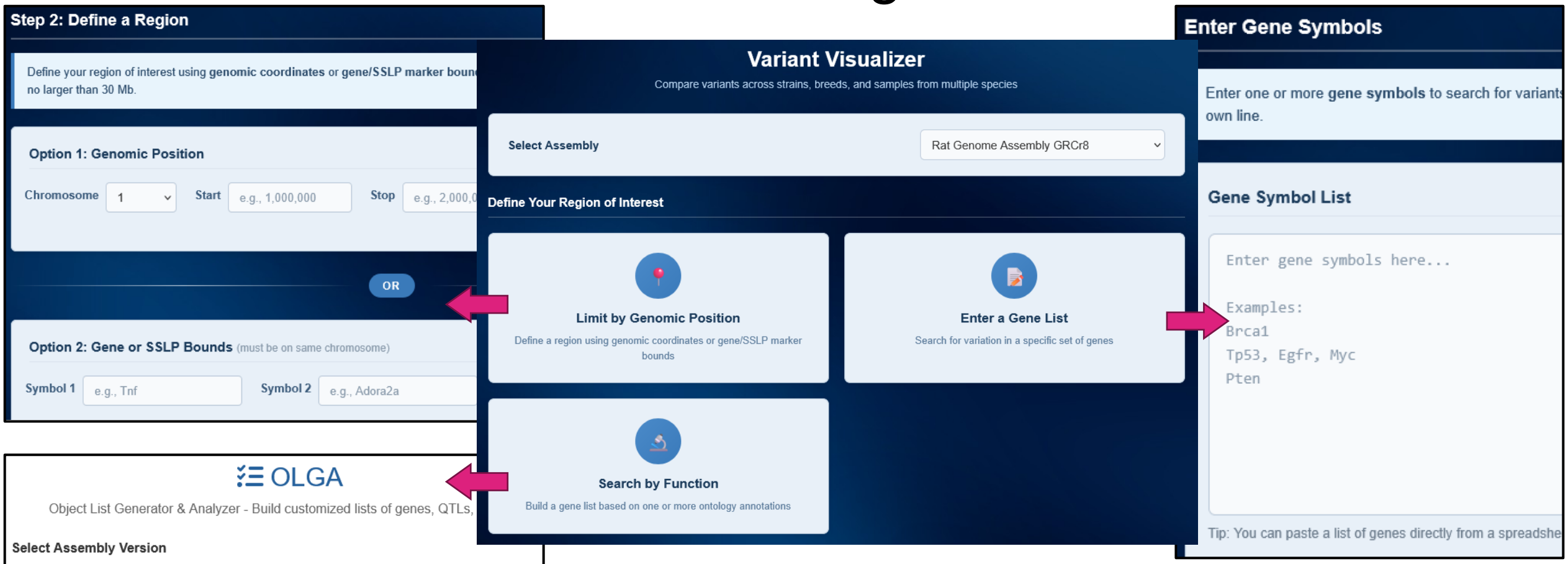
Check or uncheck individual samples to add or remove them from the selected list.

With two additional WKY samples selected the number chosen is now 10.



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RGD: The Variant Visualizer Region Selection

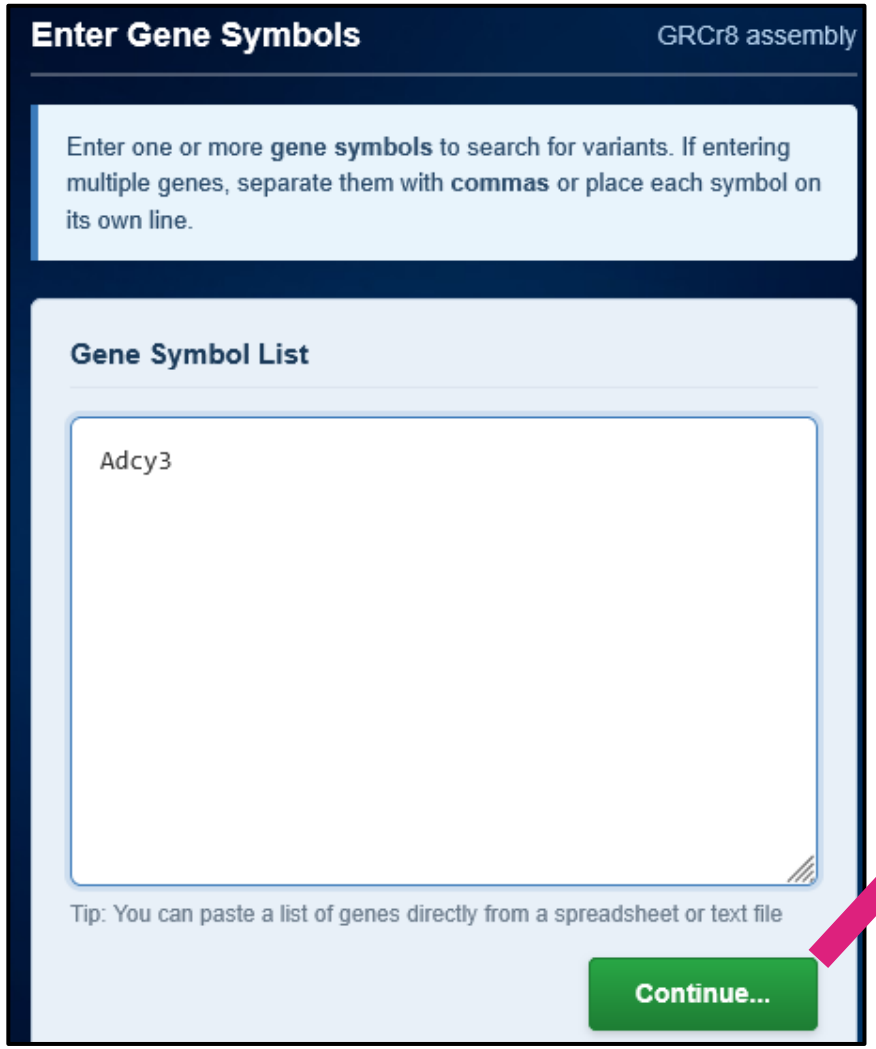


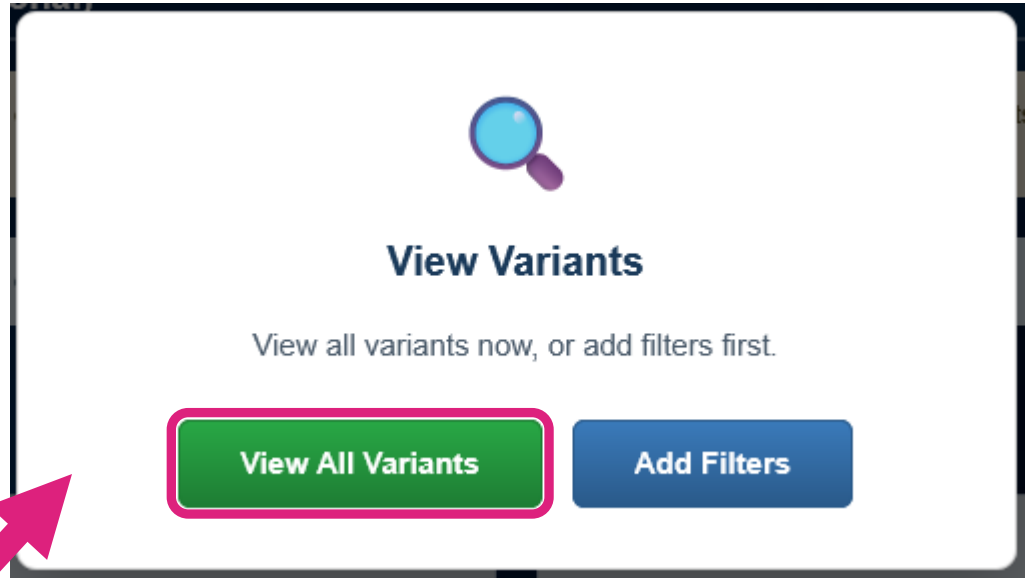
Three options are provided for defining one or more regions in which to search for variants

For this example, choose "Enter a Gene List" and type **Adcy3** in the Gene Symbol List box.

https://rgd.mcw.edu

RGD: The Variant Visualizer "View Variants"





After clicking "Continue", choose whether to view all of the variants in the selected region or gene(s) or filter the selection.

For the moment, we will choose "View All Variants" and will come back to the filter options later.

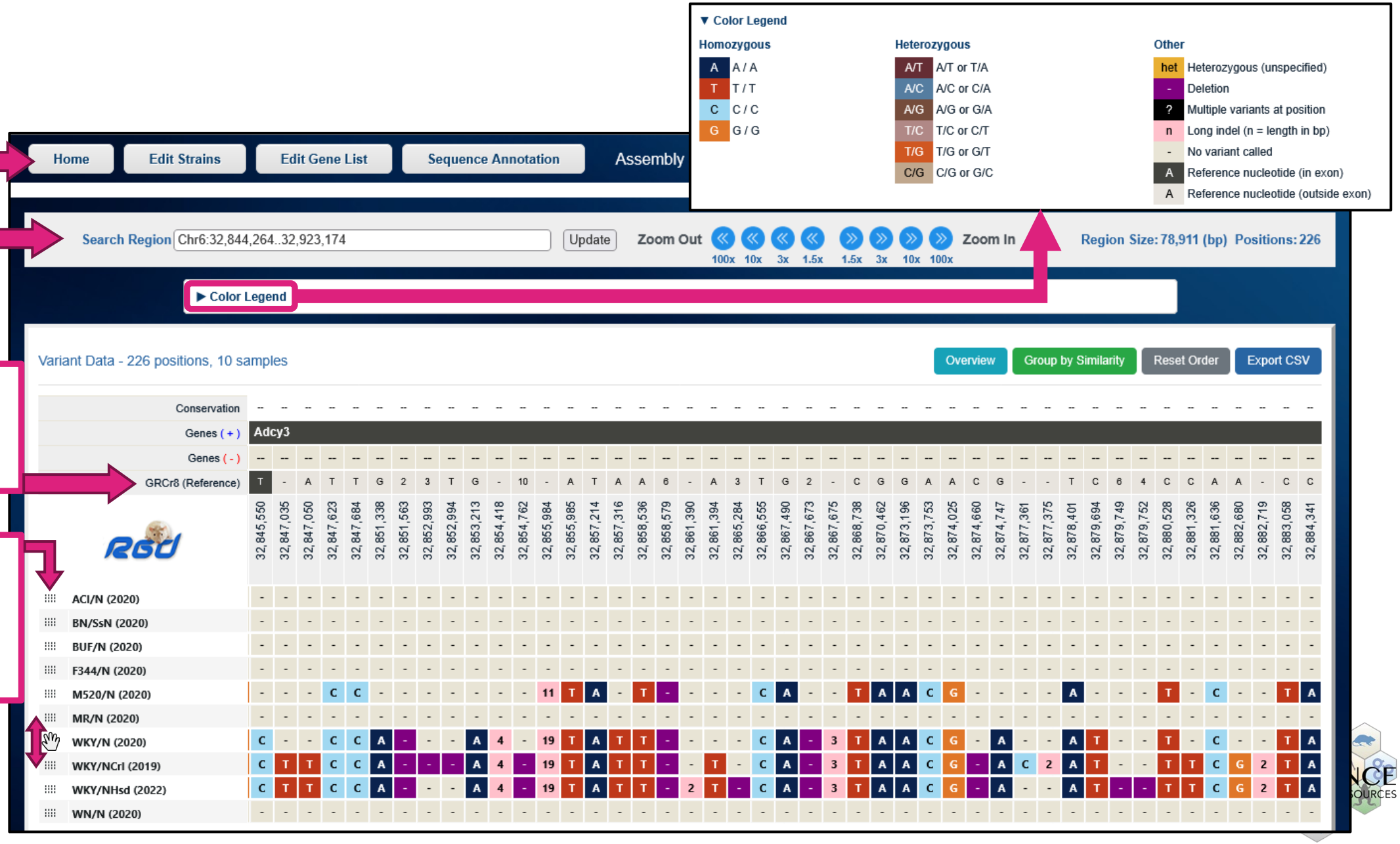
RGD: The Variant Visualizer Viewer (All Variants)

Go back to previous selection screens

Adjust the region

Shaded boxes denote exonic positions

Use "handles" to drag and drop rows to reorder strains

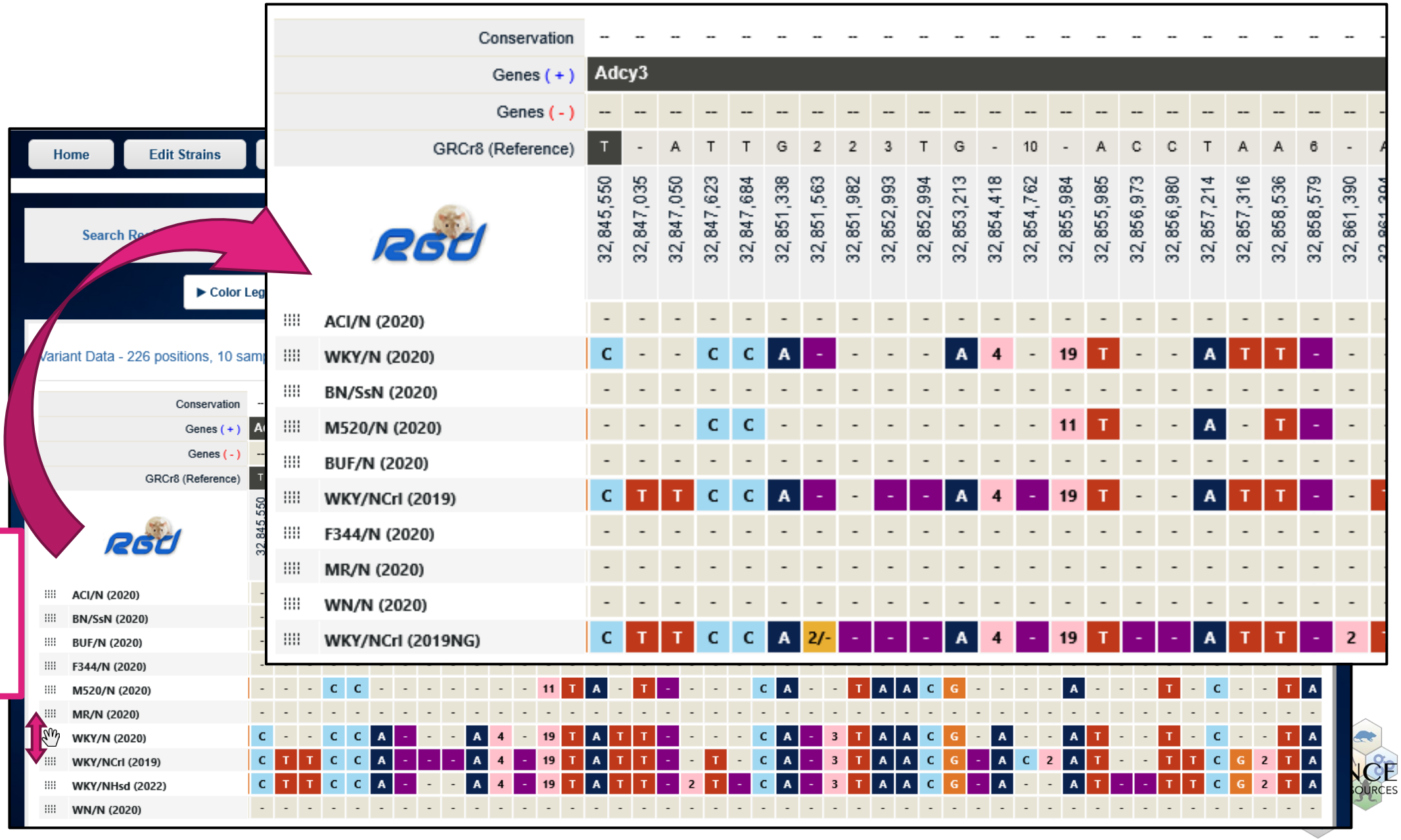


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RGD: The Variant Visualizer Viewer (All Variants)

Here the WKY samples have been dispersed throughout the list.

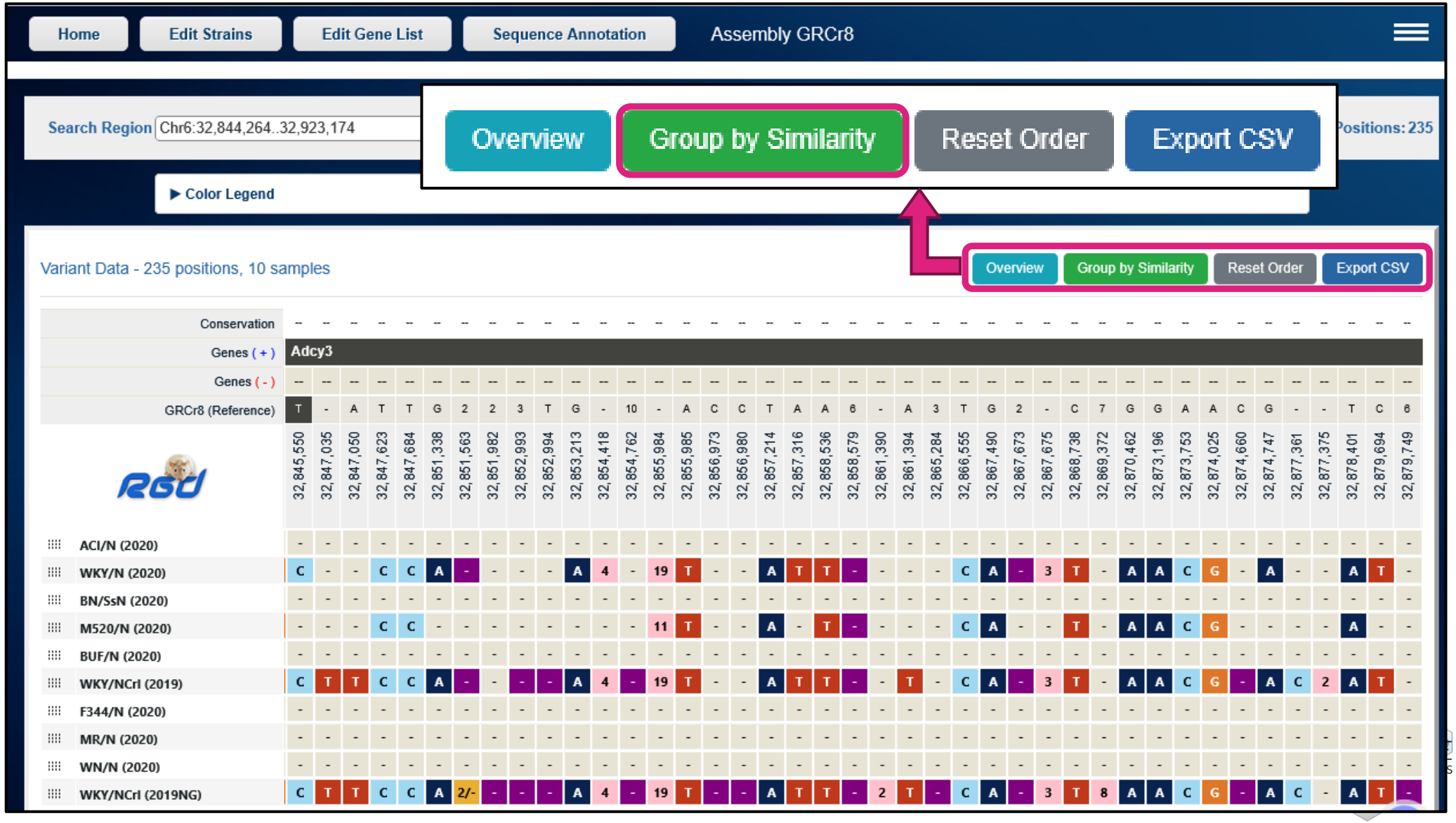
Use "handles" to drag and drop rows to reorder strains



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RGD: The Variant Visualizer Viewer (All Variants)

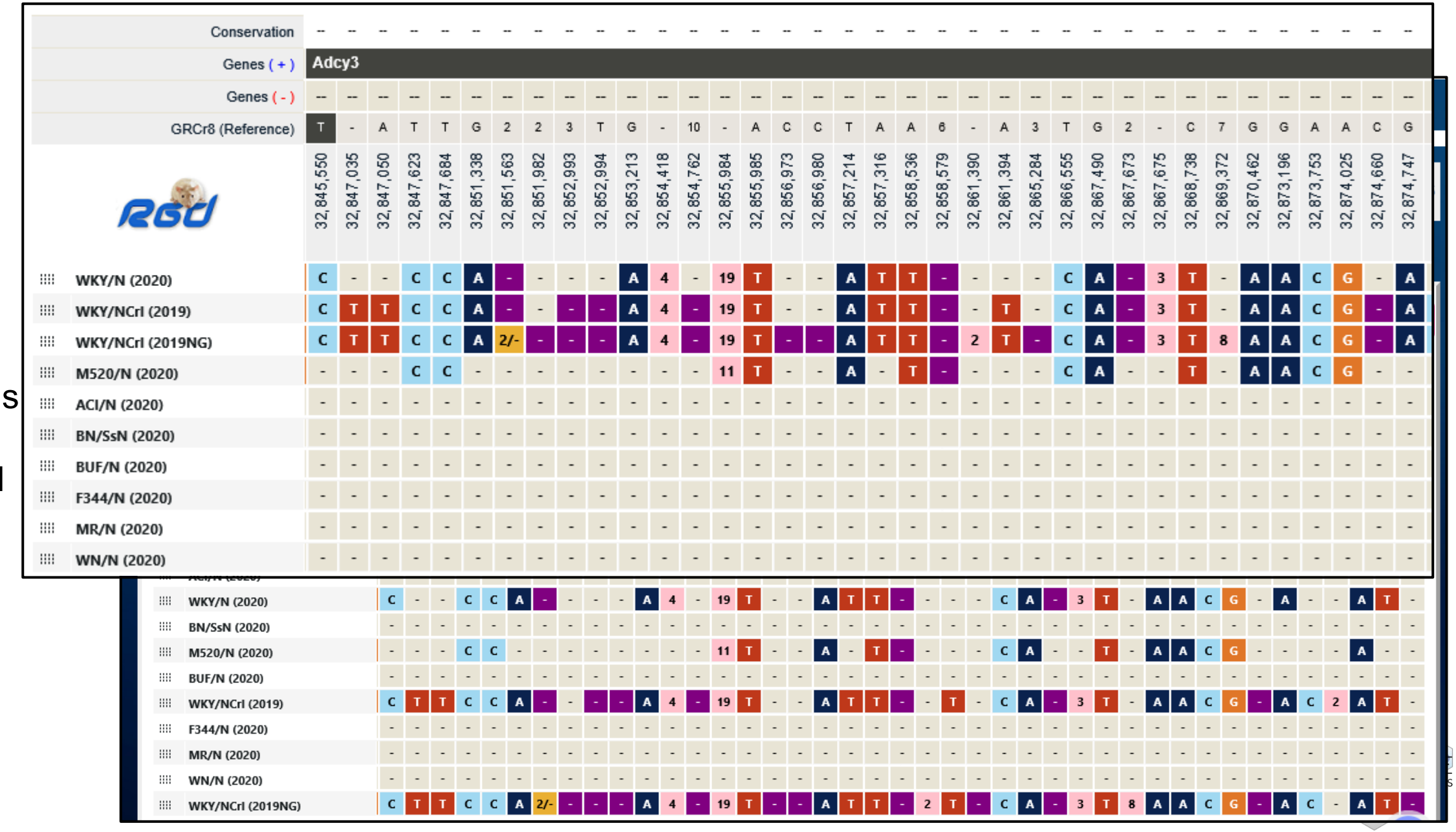
Buttons on the right above the viewer provide additional functionality, for example "Group by Similarity".



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RGD: The Variant Visualizer Viewer (All Variants)

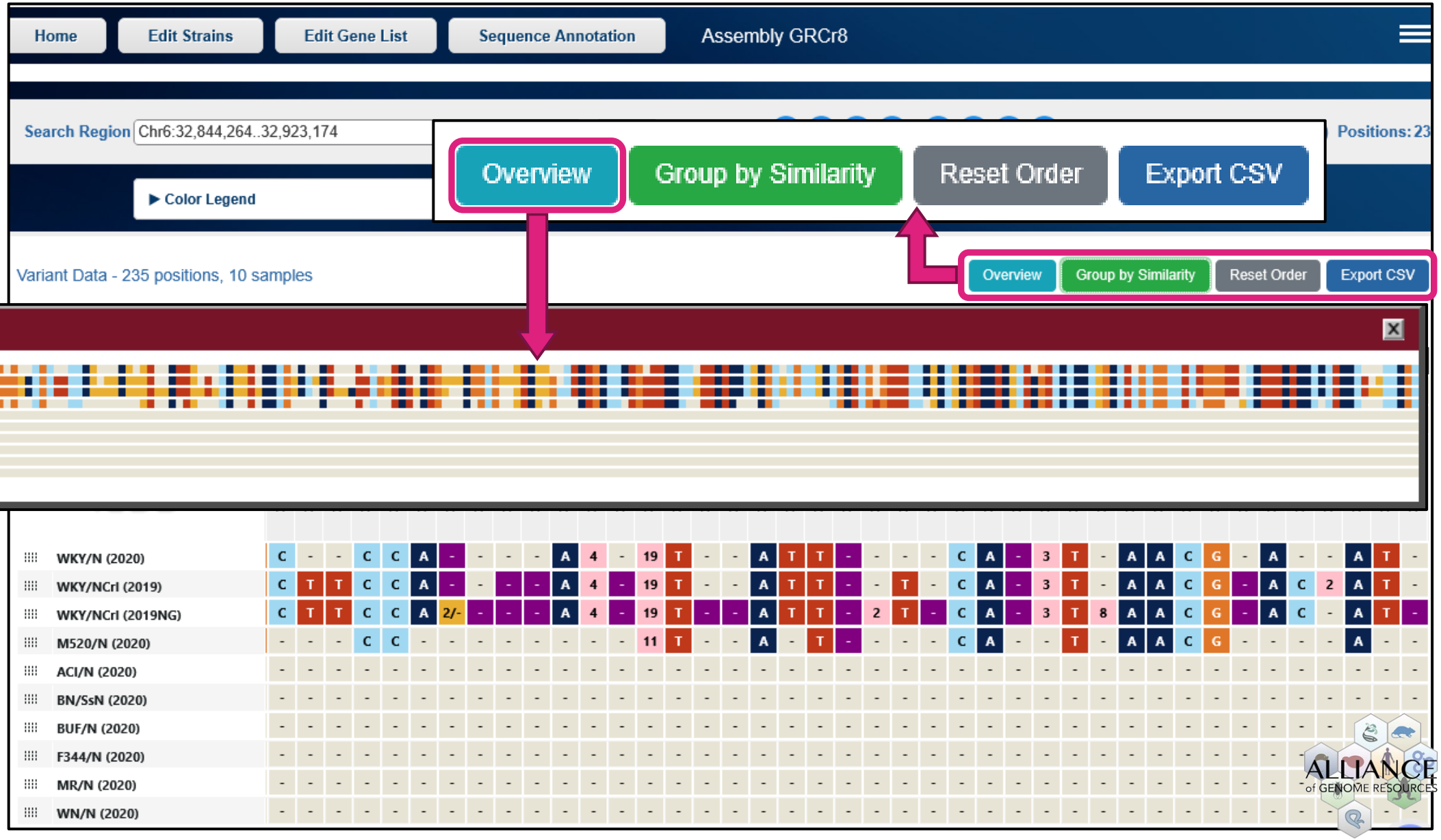
Group by Similarity has been used to group the rows with variant distributions similar to the WKY/N (2020) sample.



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RGD: The Variant Visualizer Viewer (All Variants)

"Overview" gives a bird's eye view of all of the variants in the selected region,

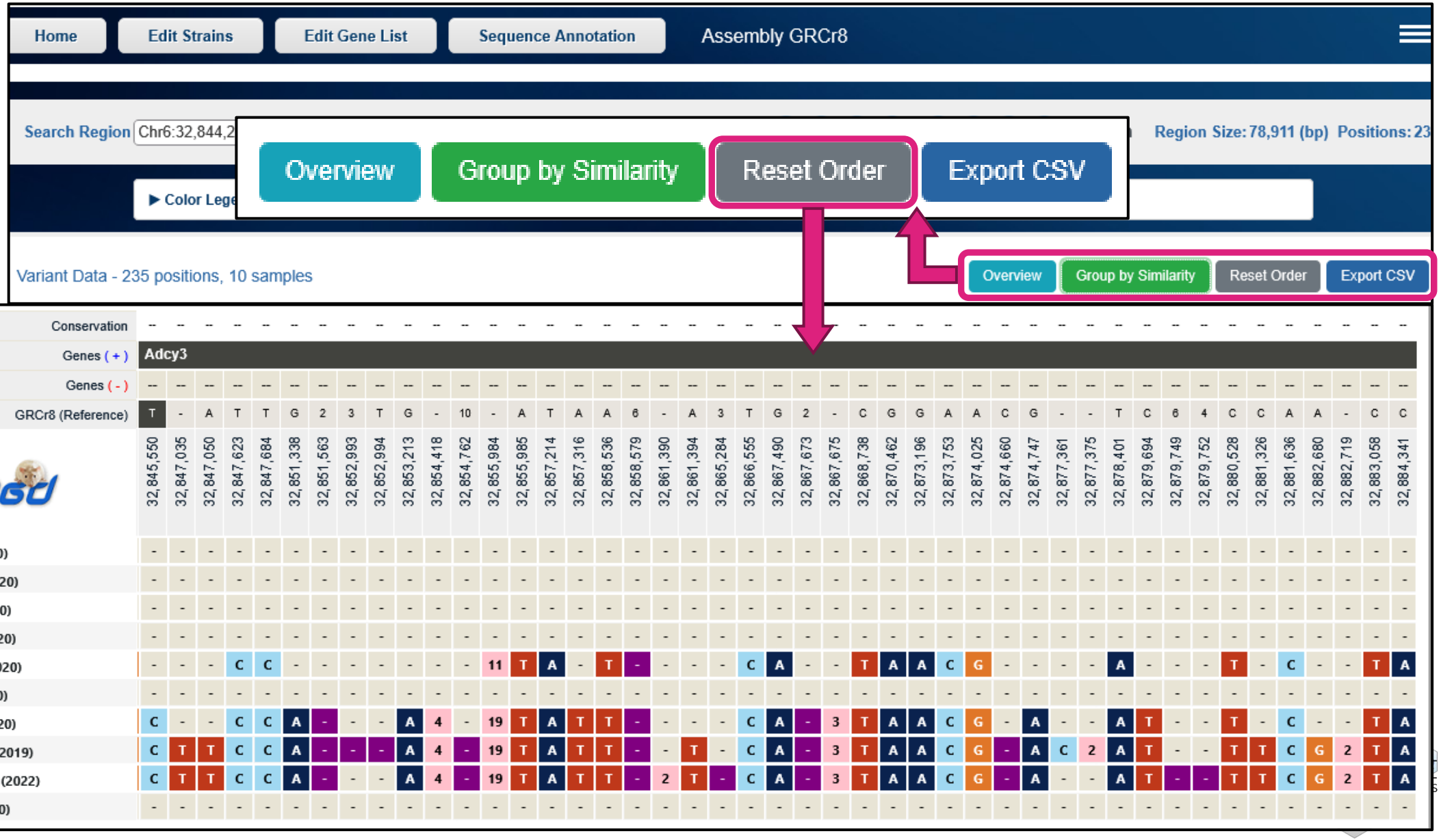


including those that are "off screen" in the main viewer.

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RGD: The Variant Visualizer Viewer (All Variants)

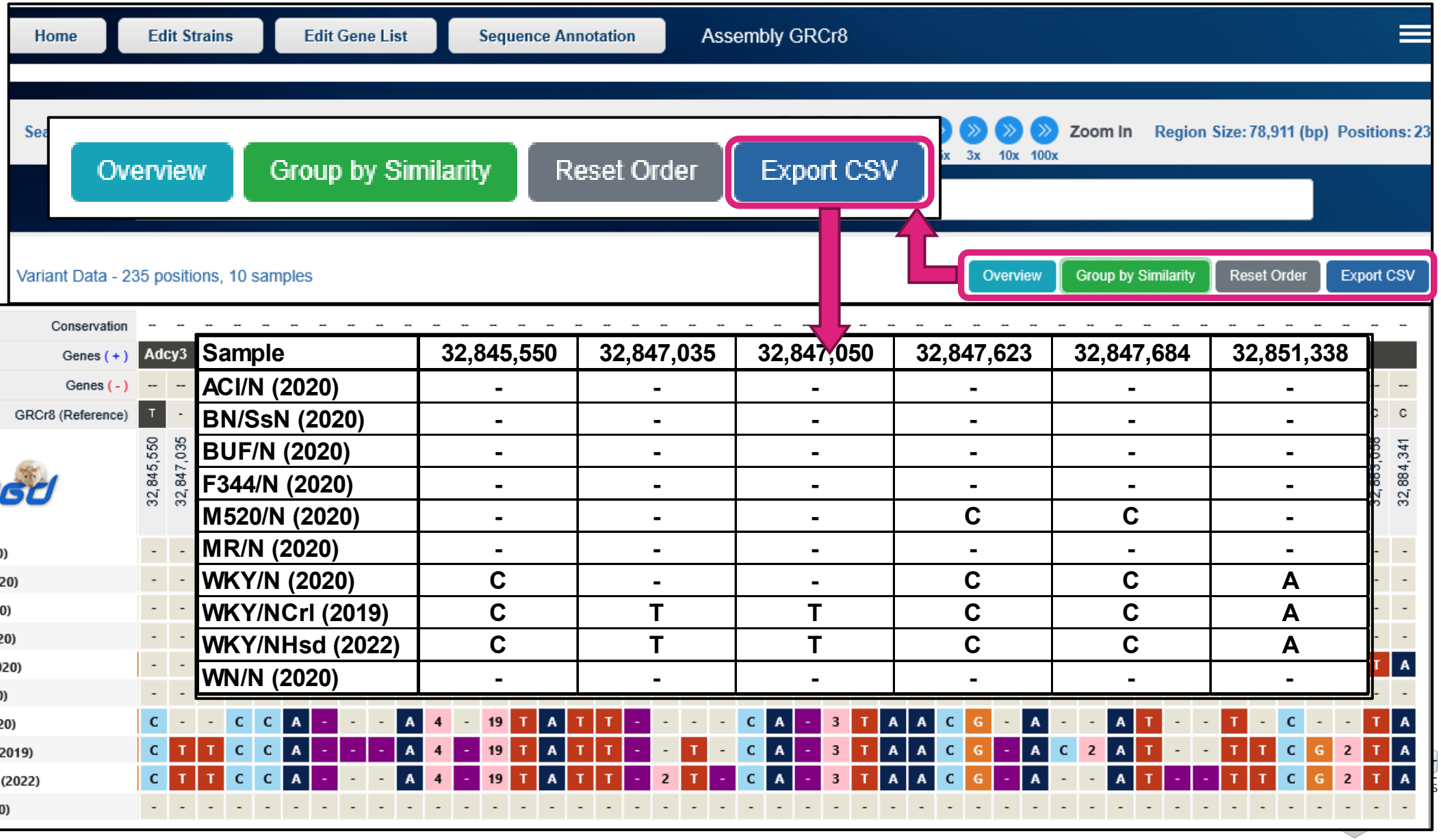
"Reset Order" returns the order of the rows to alphanumeric by sample name.



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RGD: The Variant Visualizer Viewer (All Variants)

"Export CSV" downloads a CSV file that recapitulates the view of the variants. All variants in the region are included.

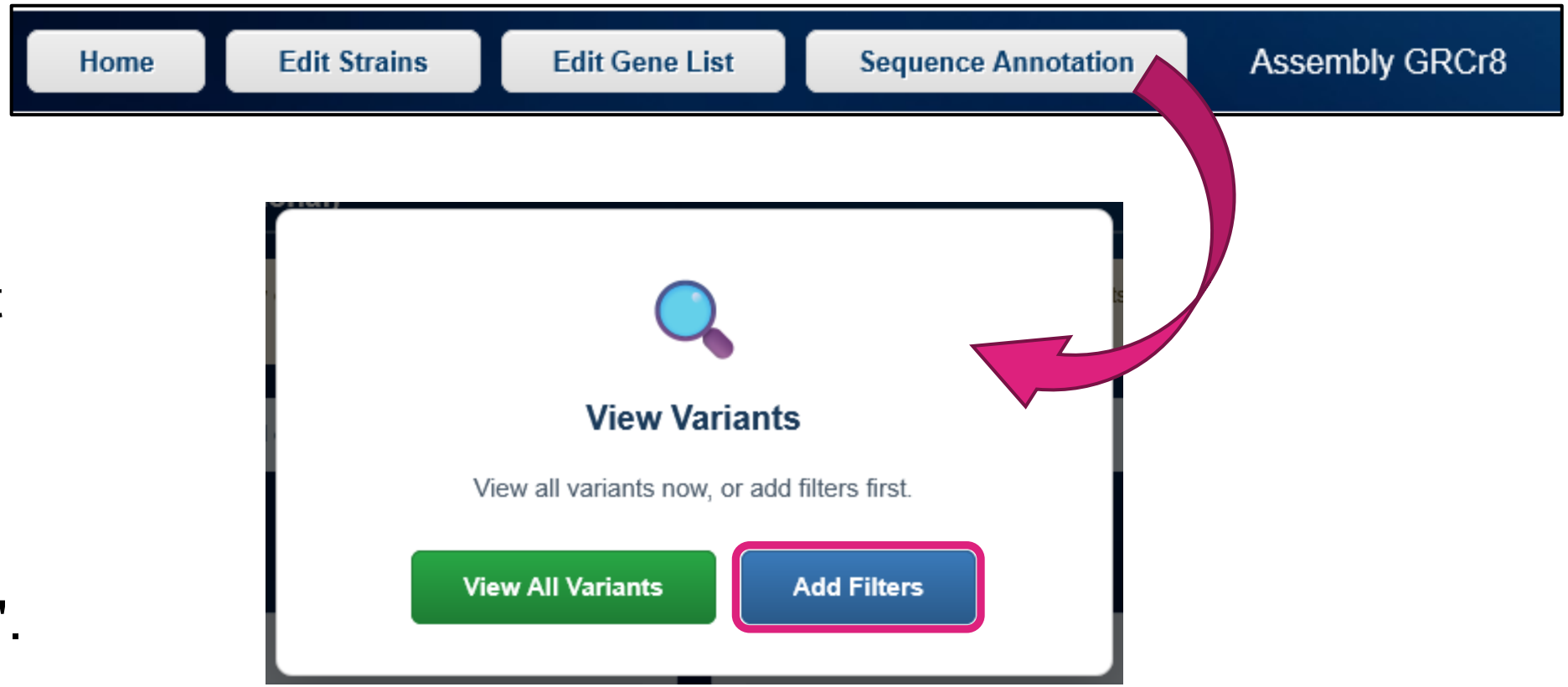


Sample	32,845,550	32,847,035	32,847,050	32,847,623	32,847,684	32,851,338
ACI/N (2020)	-	-	-	-	-	-
BN/SsN (2020)	-	-	-	-	-	-
BUF/N (2020)	-	-	-	-	-	-
F344/N (2020)	-	-	-	-	-	-
M520/N (2020)	-	-	-	C	C	-
MR/N (2020)	-	-	-	-	-	-
WKY/N (2020)	C	-	-	C	C	A
WKY/NCrl (2019)	C	T	T	C	C	A
WKY/NHsd (2022)	C	T	T	C	C	A
WN/N (2020)	-	-	-	-	-	-

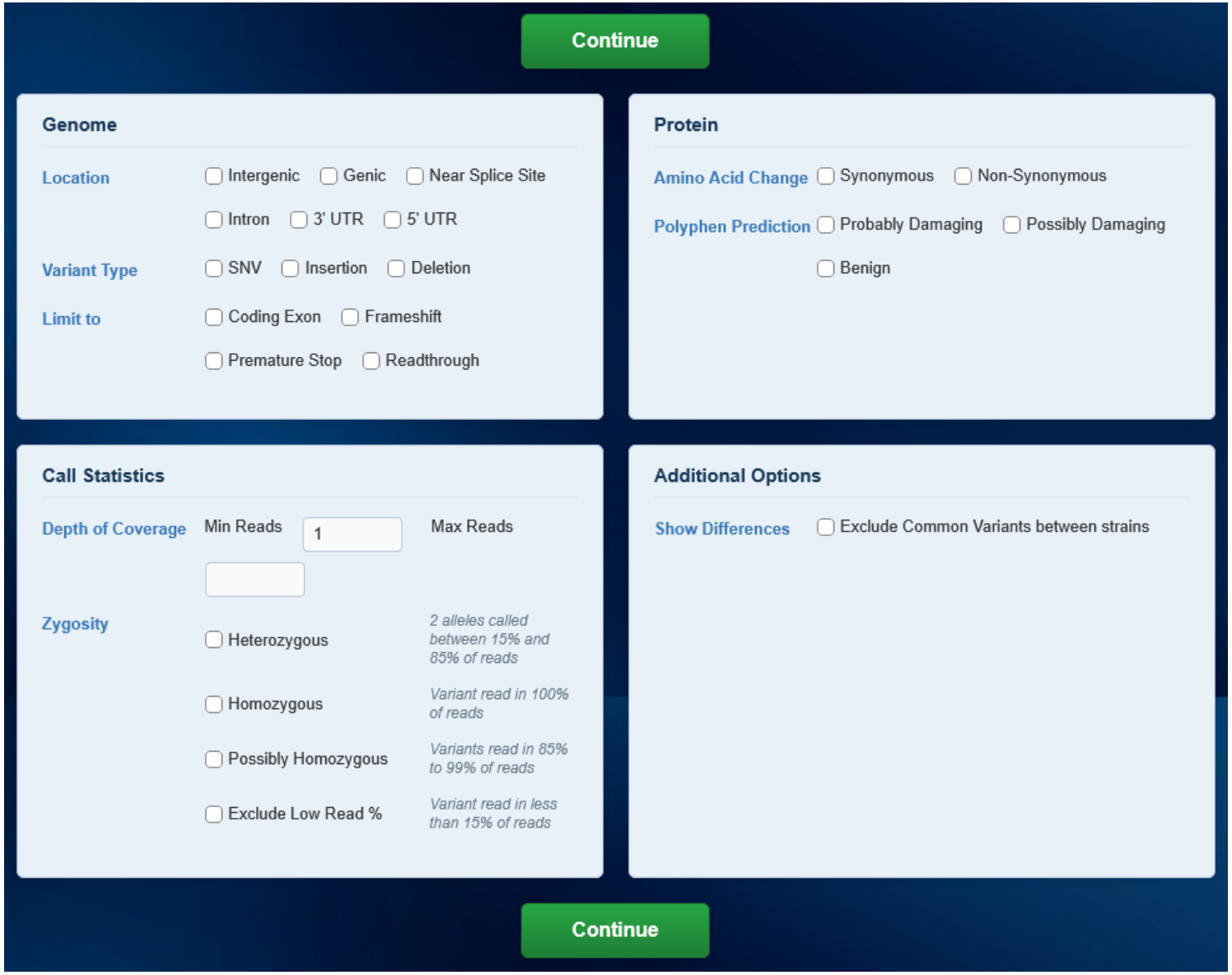
https://rgd.mcw.edu

RGD: The Variant Visualizer Viewer (Filtering Variants)

In the Variant Visualizer Viewer, click on the "Sequence Annotation" button at the top of the viewer window. This will once again bring up the selector to "View All…" or "Add Filters". This time, click "Add Filters".

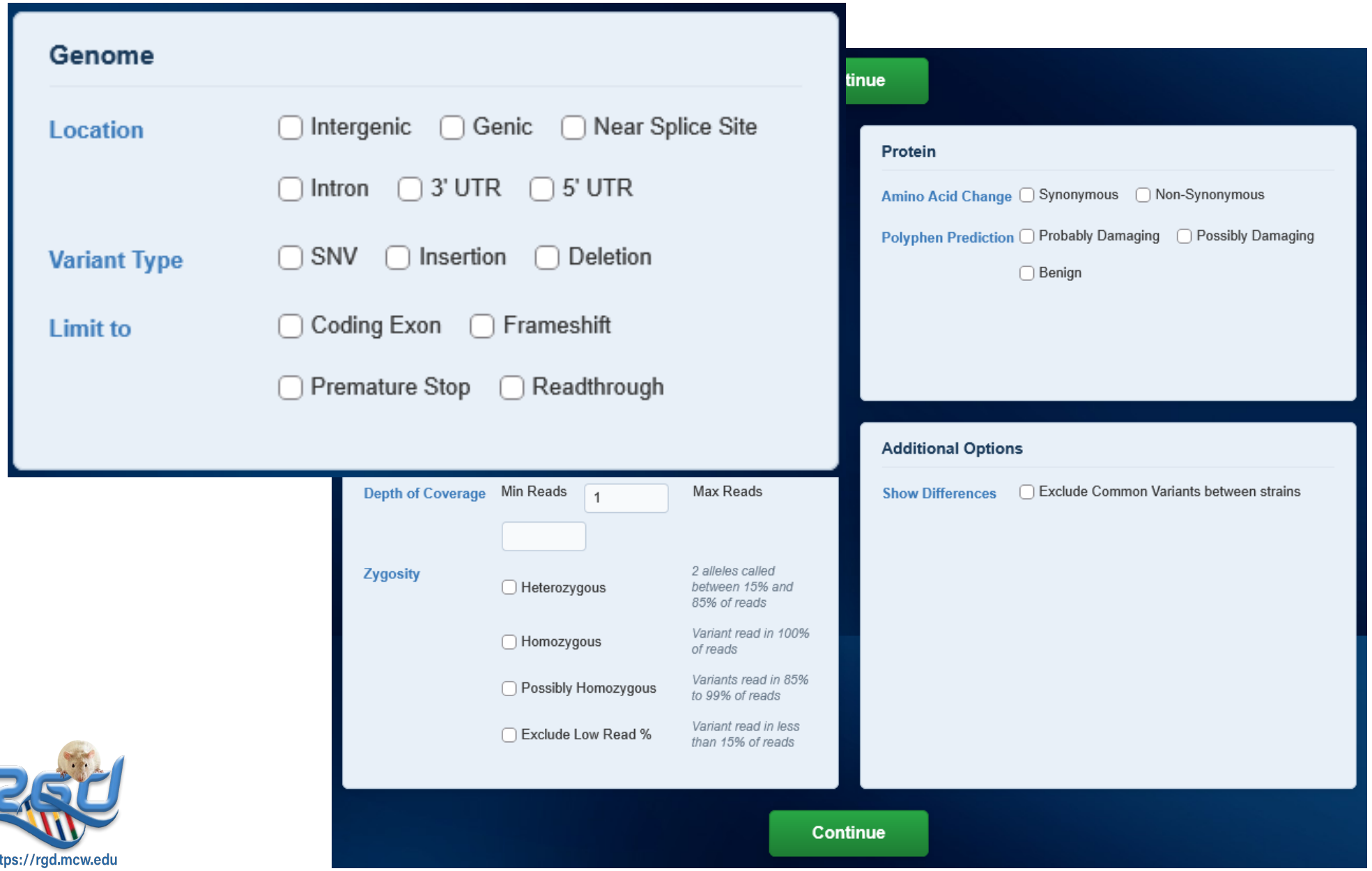


RGD: The Variant Visualizer Viewer (Filtering Variants)



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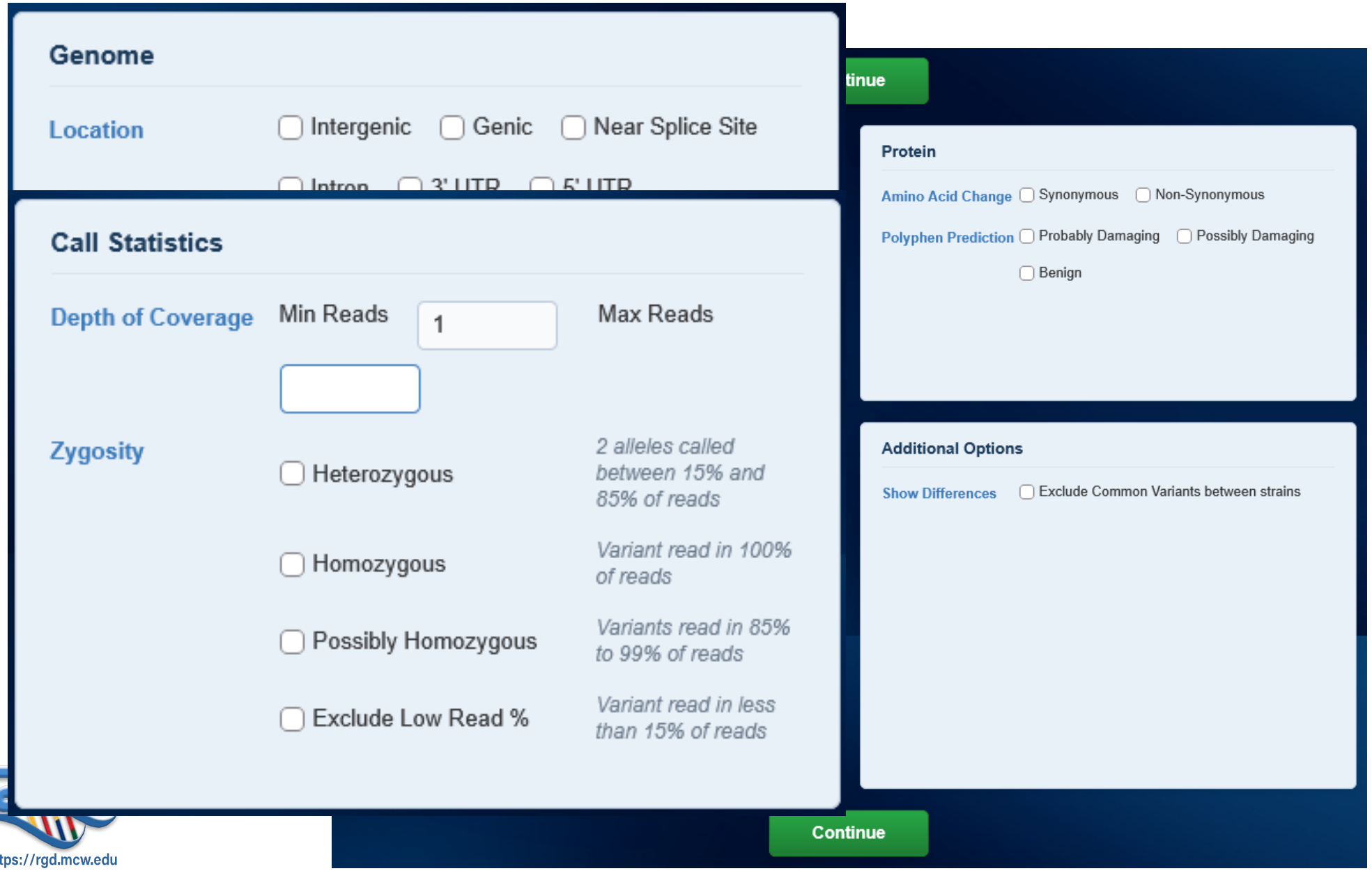
RGD: The Variant Visualizer Viewer (Filtering Variants)



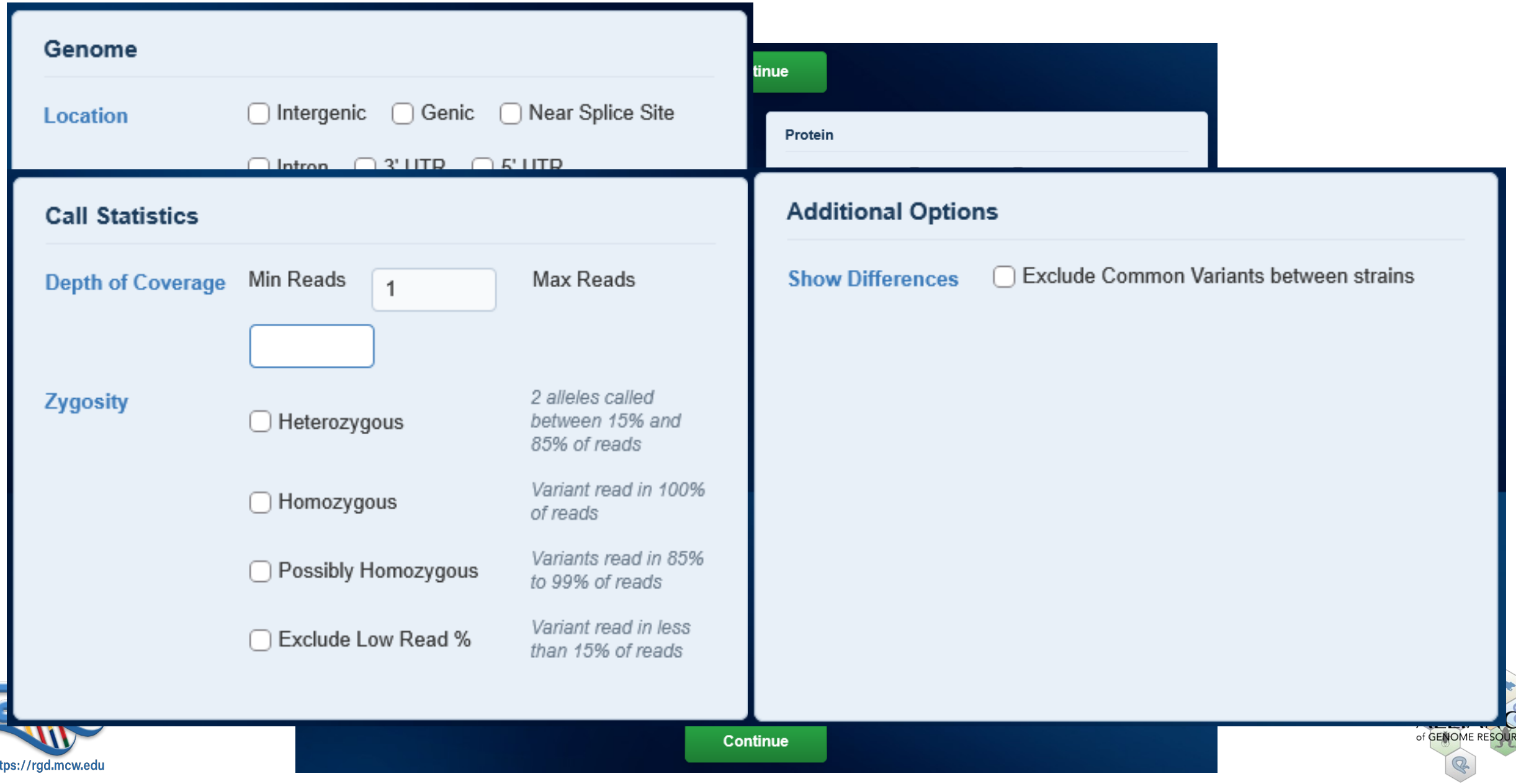
https://rgd.mcw.edu

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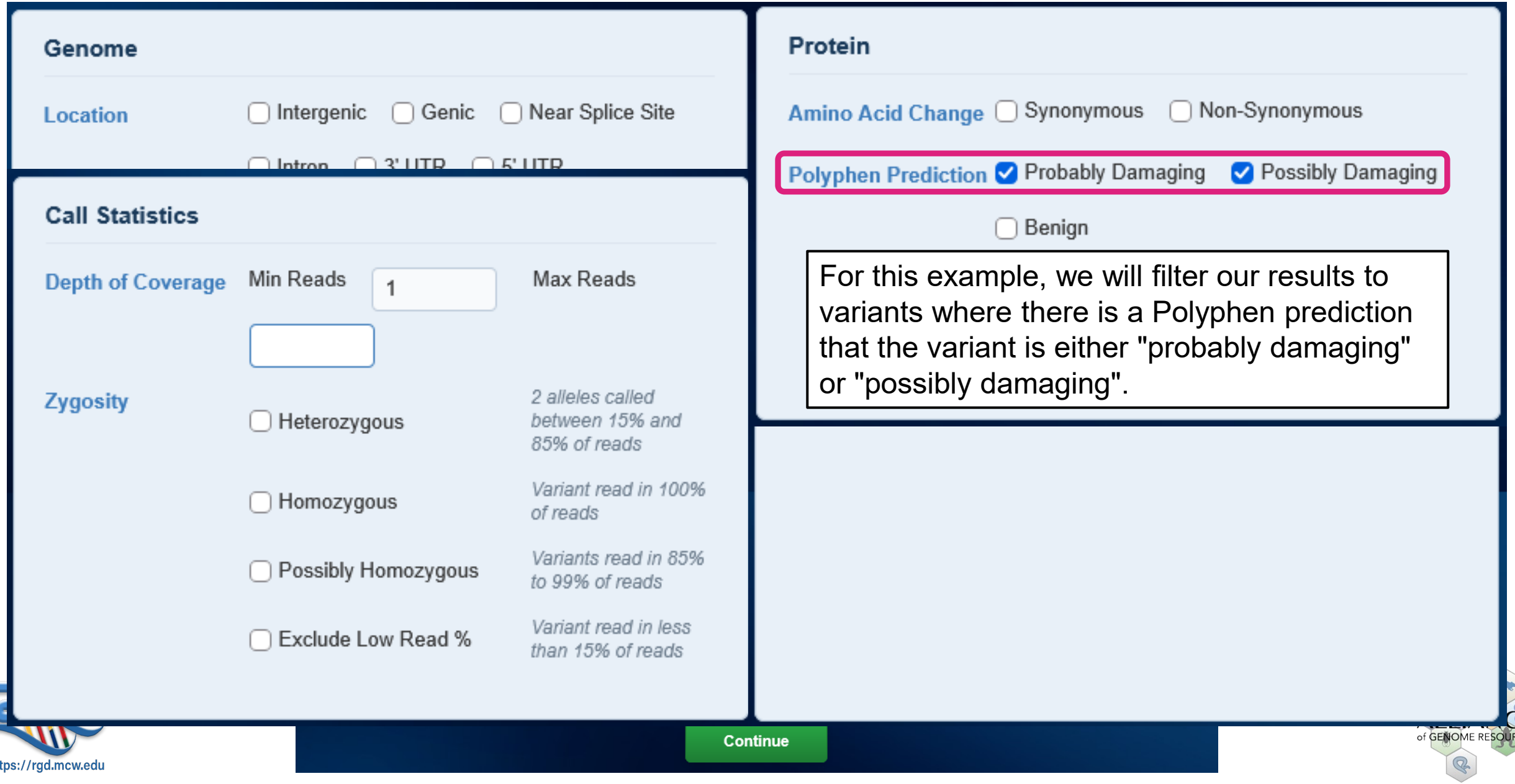
RGD: The Variant Visualizer Viewer (Filtering Variants)



RGD: The Variant Visualizer Viewer (Filtering Variants)

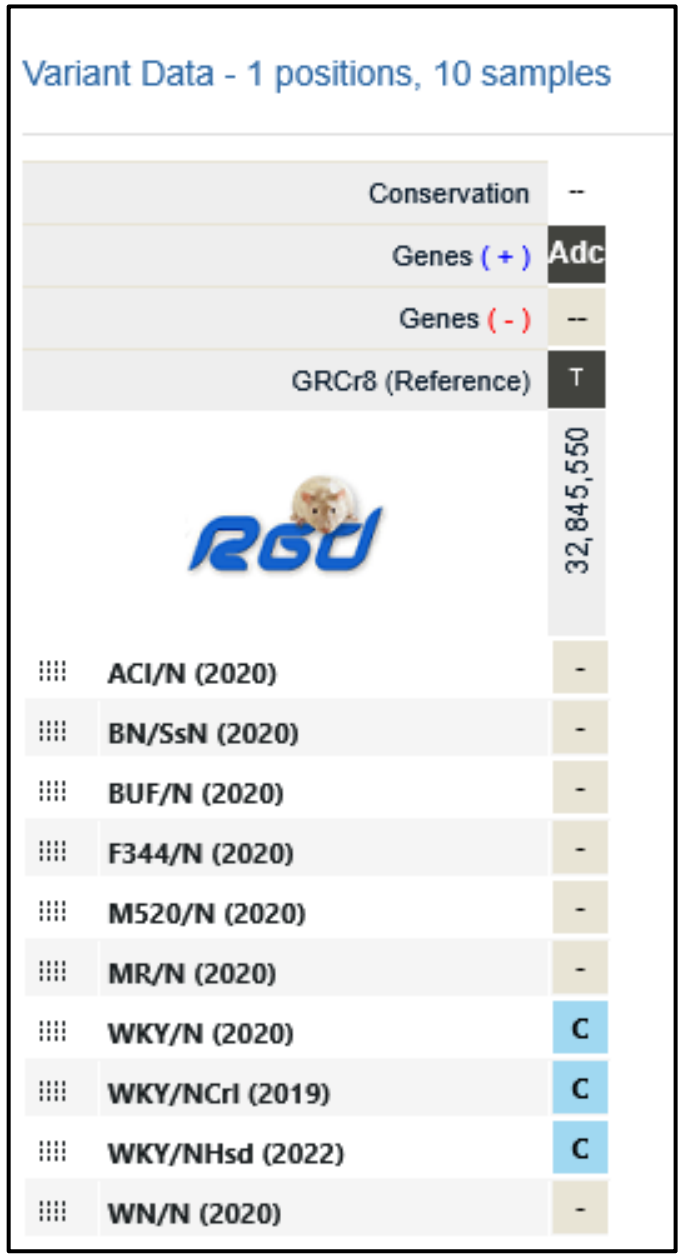


RGD: The Variant Visualizer Viewer (Filtering Variants)



RGD: The Variant Visualizer Viewer (Filtering Variants)

After filtering for only possibly and probably damaging variants, a single T-to-C variant found in the three WKY samples remains.

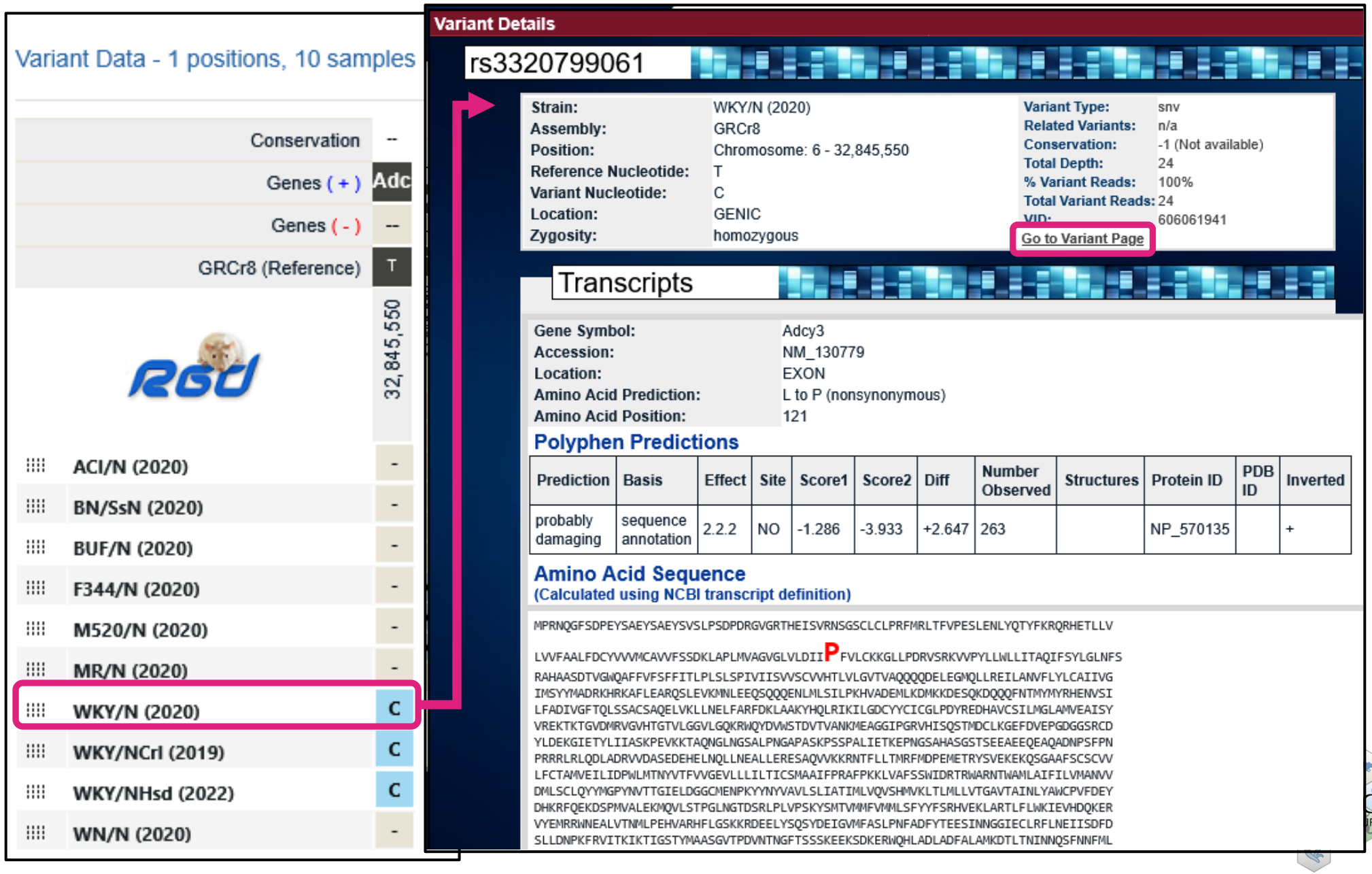


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RGD: The Variant Visualizer Viewer (Filtering Variants)

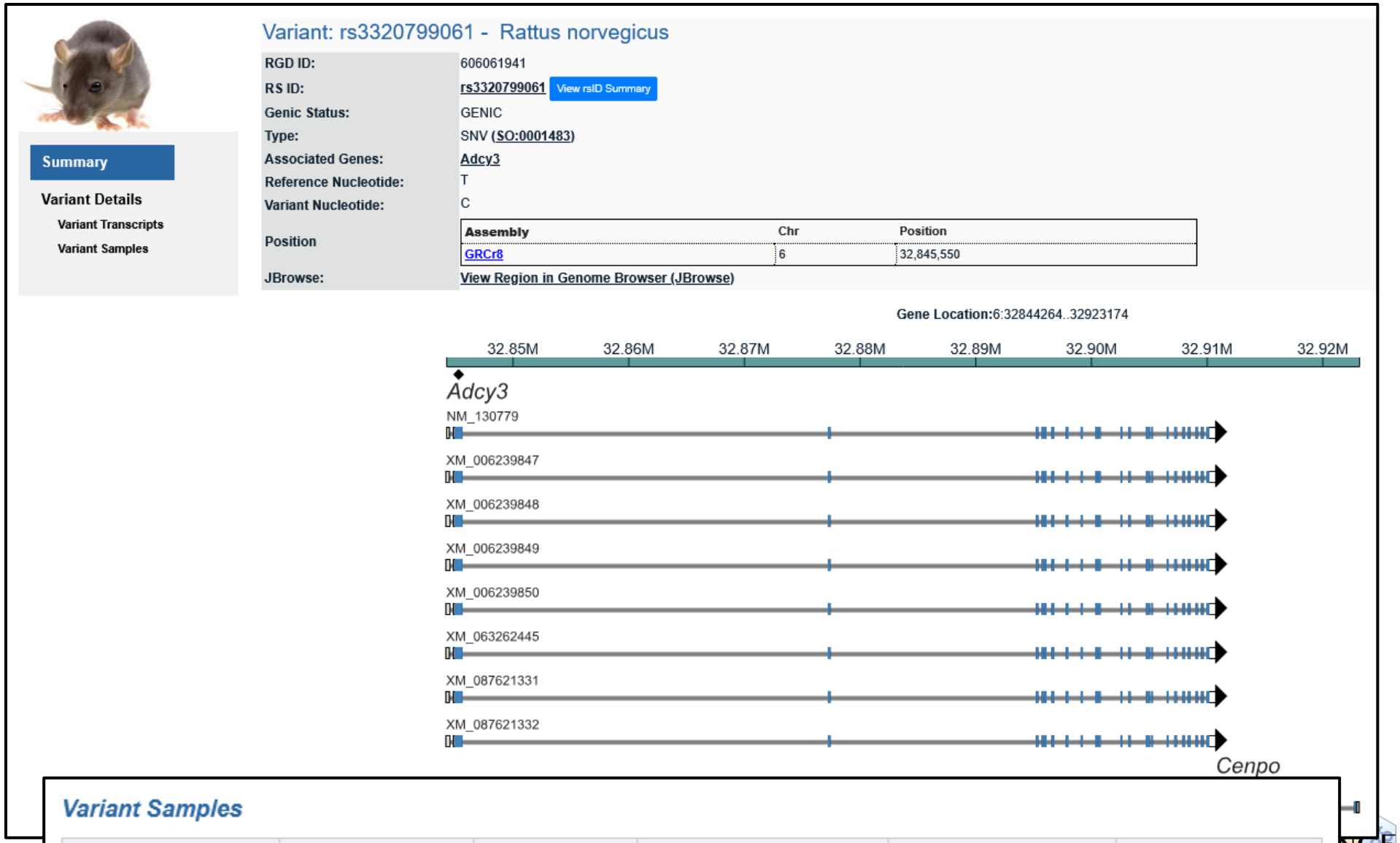
Click on a square in the viewer to bring up a Variant Details popup with information about the variant and its effects.



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RGD: The Variant Report Page

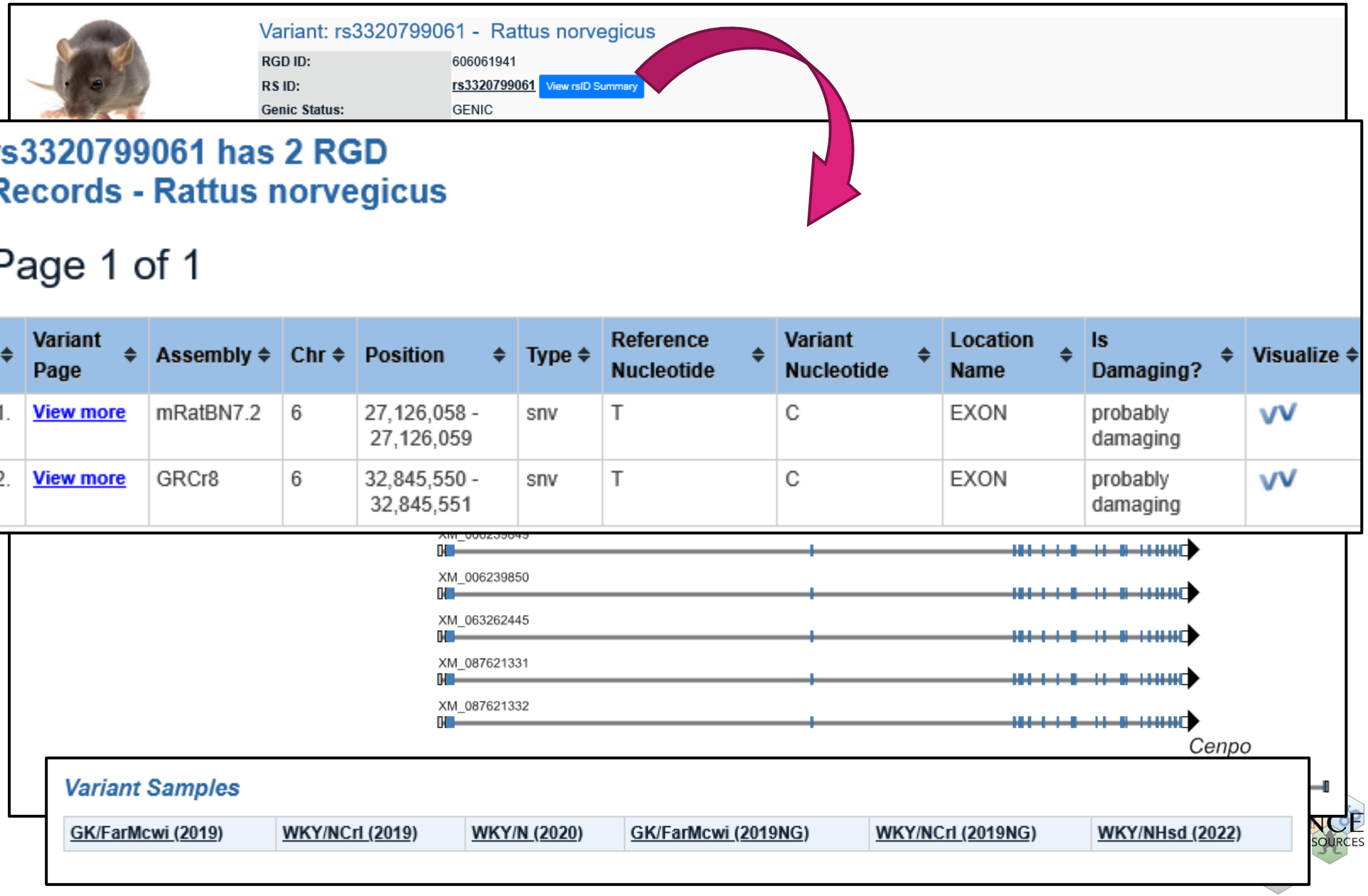
The Variant Details popup links to the RGD variant report page for that variant as shown here.



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RGD: The Variant Report Page

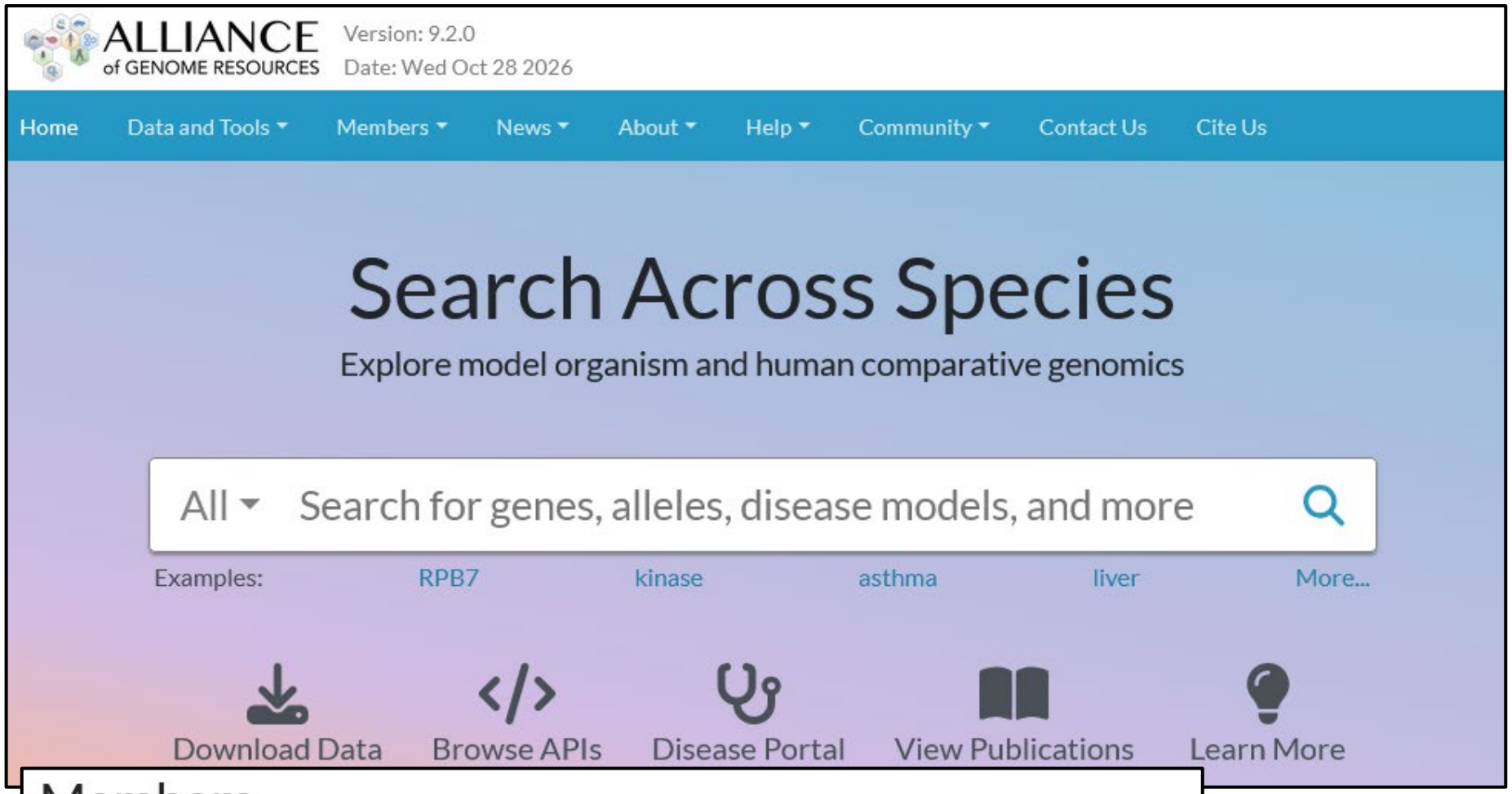
RGD has submitted the strain-specific variants from our sequence analyses to the European Variation Archive (EVA) and imported the assigned rs IDs from them. Using the rs ID, variants can be tracked across assemblies. For example, this variant was seen in both the mRatBN7.2 assembly and GRCr8.

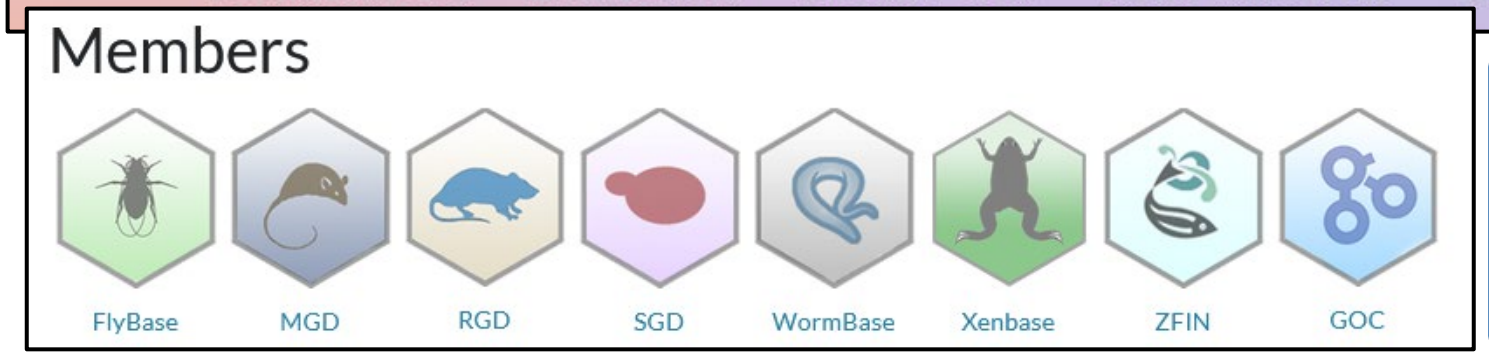


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The Alliance of Genome Resources

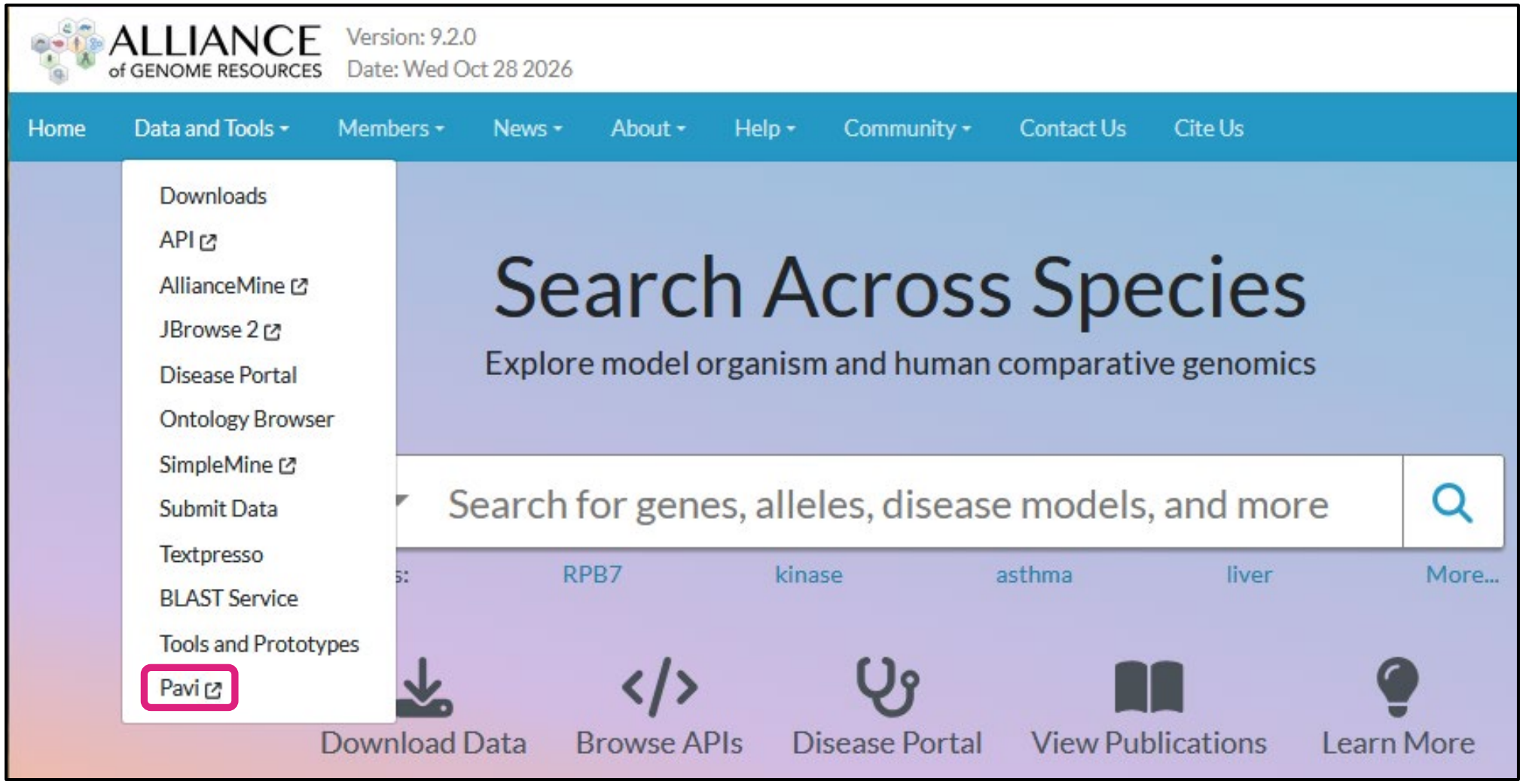
The Alliance of Genome Resources (the Alliance) is "a consortium of seven model organism databases and the Gene Ontology (GO) Consortium whose goal is to provide an integrated view of their data".





https://rgd.mcw.edu

The Alliance: the Protein Annotation and Variant Inspector



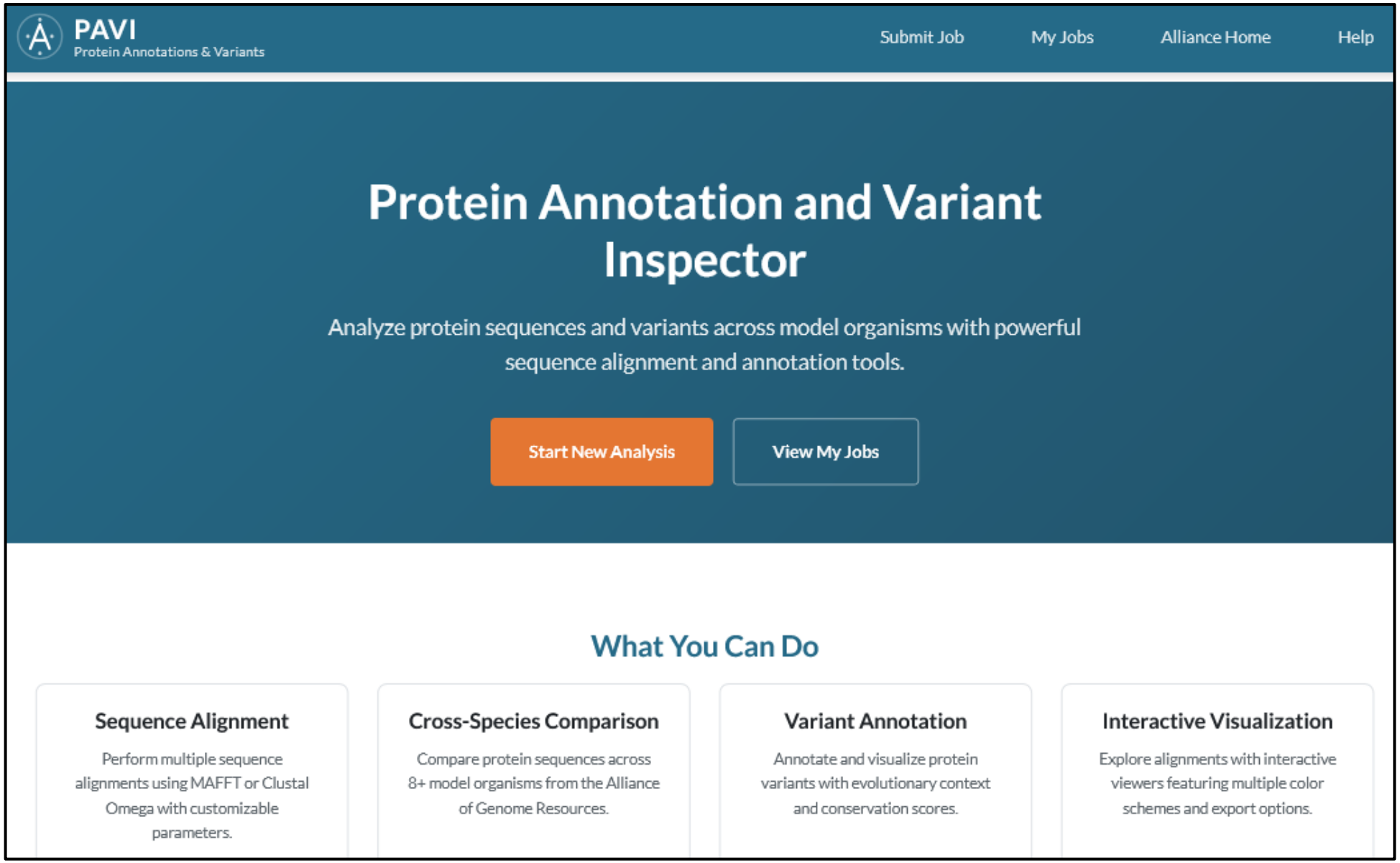
Access PAVI, the Protein Annotation and Variant Inspector tool using the "Data and Tools" menu on any Alliance page.

https://rgd.mcw.edu

The Alliance: the Protein Annotation and Variant Inspector

Access PAVI using this QR code or the URL below.

PAVI



https://rgd.mcw.edu

https://www.alliancegenome.org/pavi/

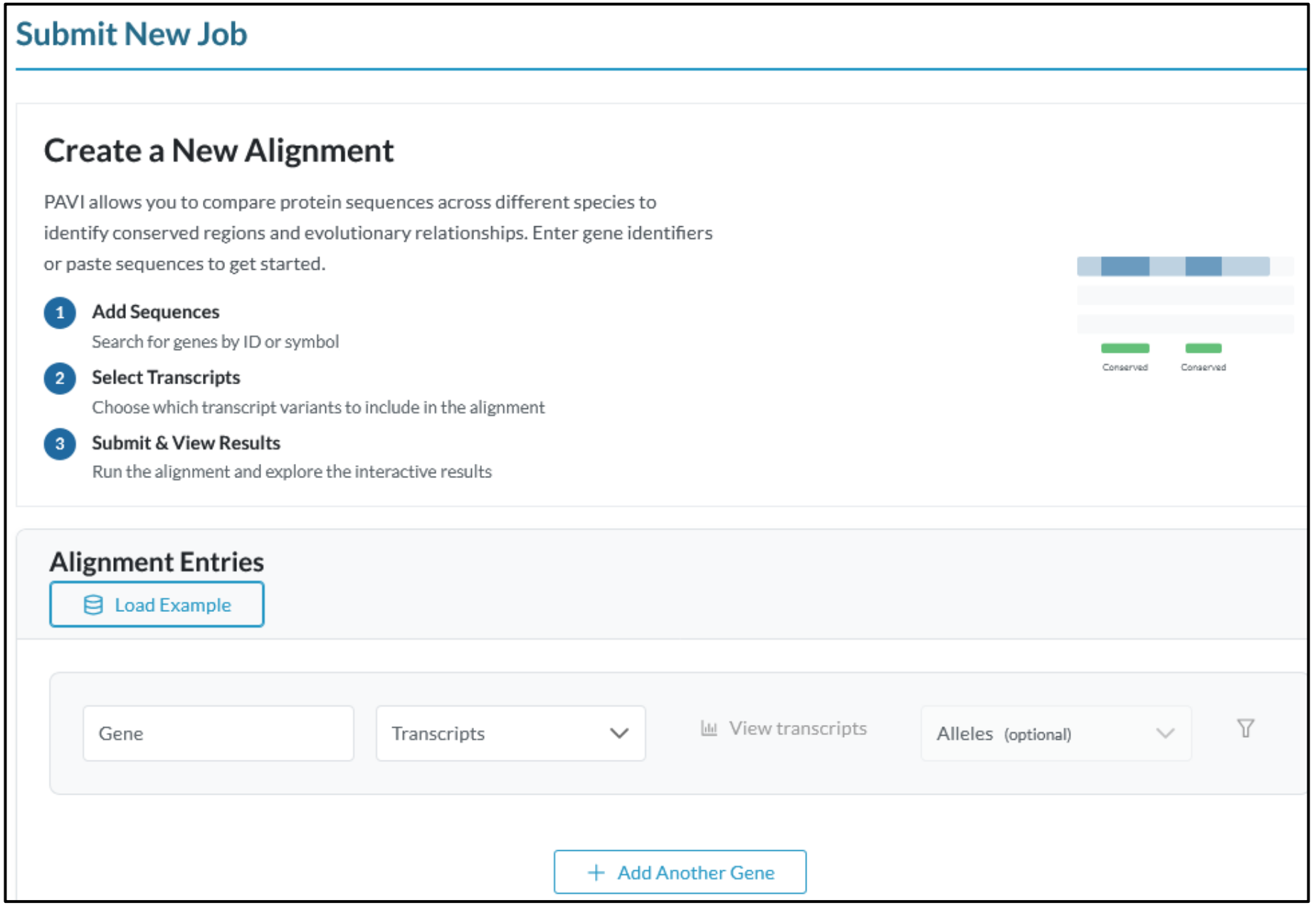
Access the recent webinar about PAVI using this QR code

About PAVI

ALLIANCE of GENOME RESOURCES

The Alliance: the Protein Annotation and Variant Inspector

Clicking "Start New Analysis" on the PAVI home page opens the "Create a New Alignment" page.

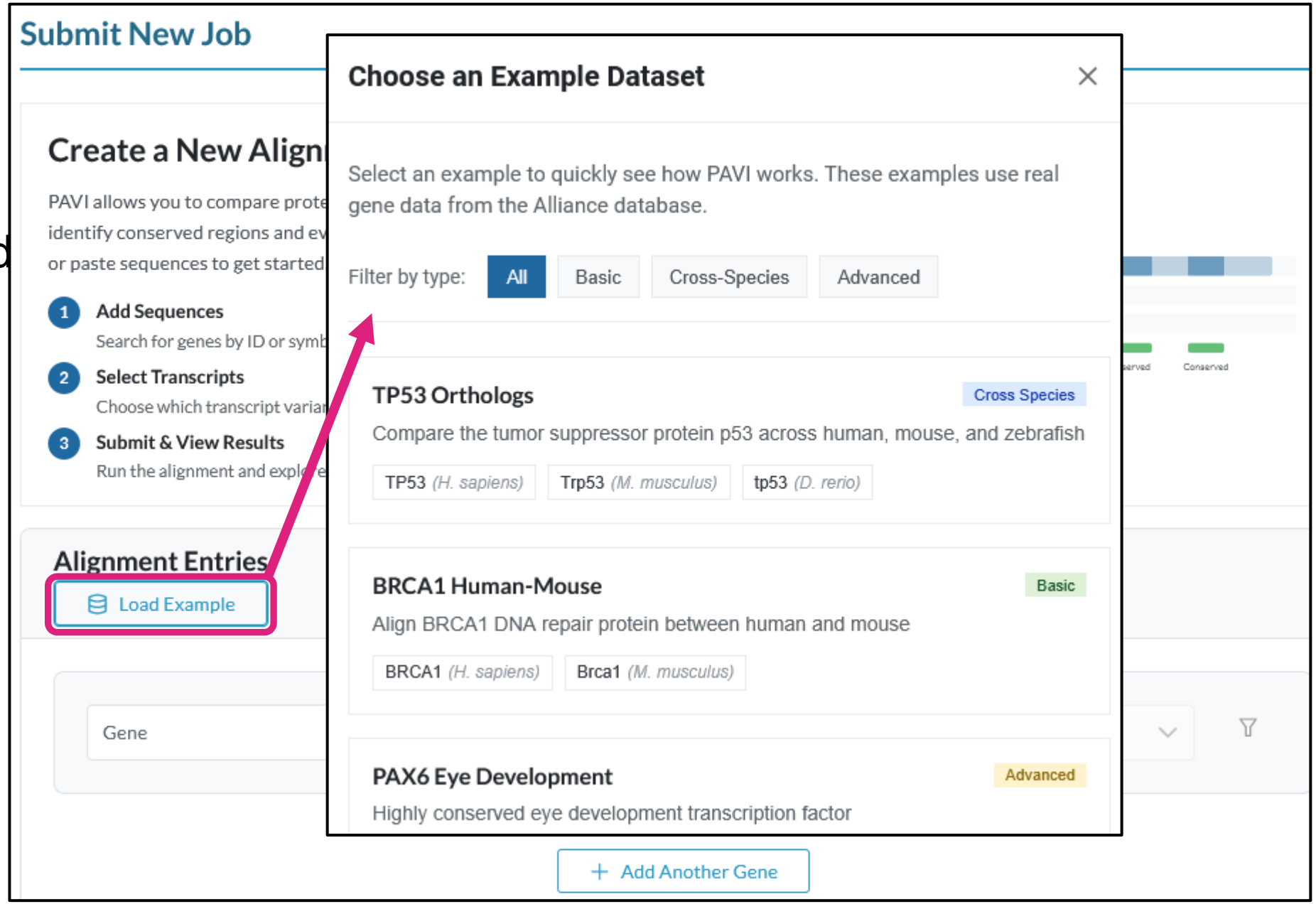


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ALLIANCE of GENOME RESOURCES

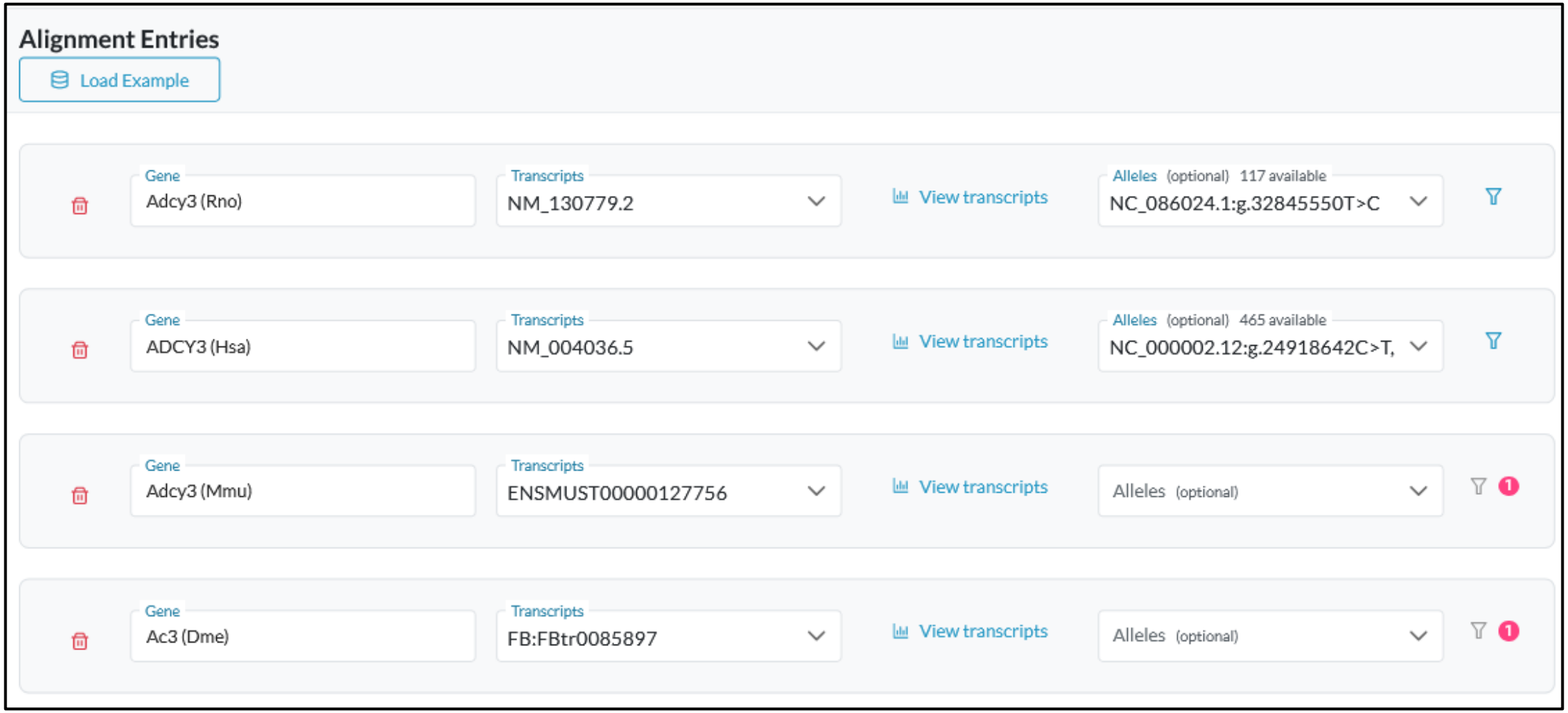
The Alliance: the Protein Annotation and Variant Inspector

To easily see how PAVI works, choose a preconfigured example dataset from the list provided.



https://rgd.mcw.edu

The Alliance: PAVI Alignment Entries



This is the entry page for the example that will be shown in later slides. The values, presented in a format that can be copied and pasted into the tool, are on the next slide.

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The Alliance: PAVI Alignment Entries

The input values to create the same view as the example are below. Use any or all of these to try the tool yourself. For reference and as will be shown in the coming slides, the variant for rat and the third variant for human are at the same location in the alignment.

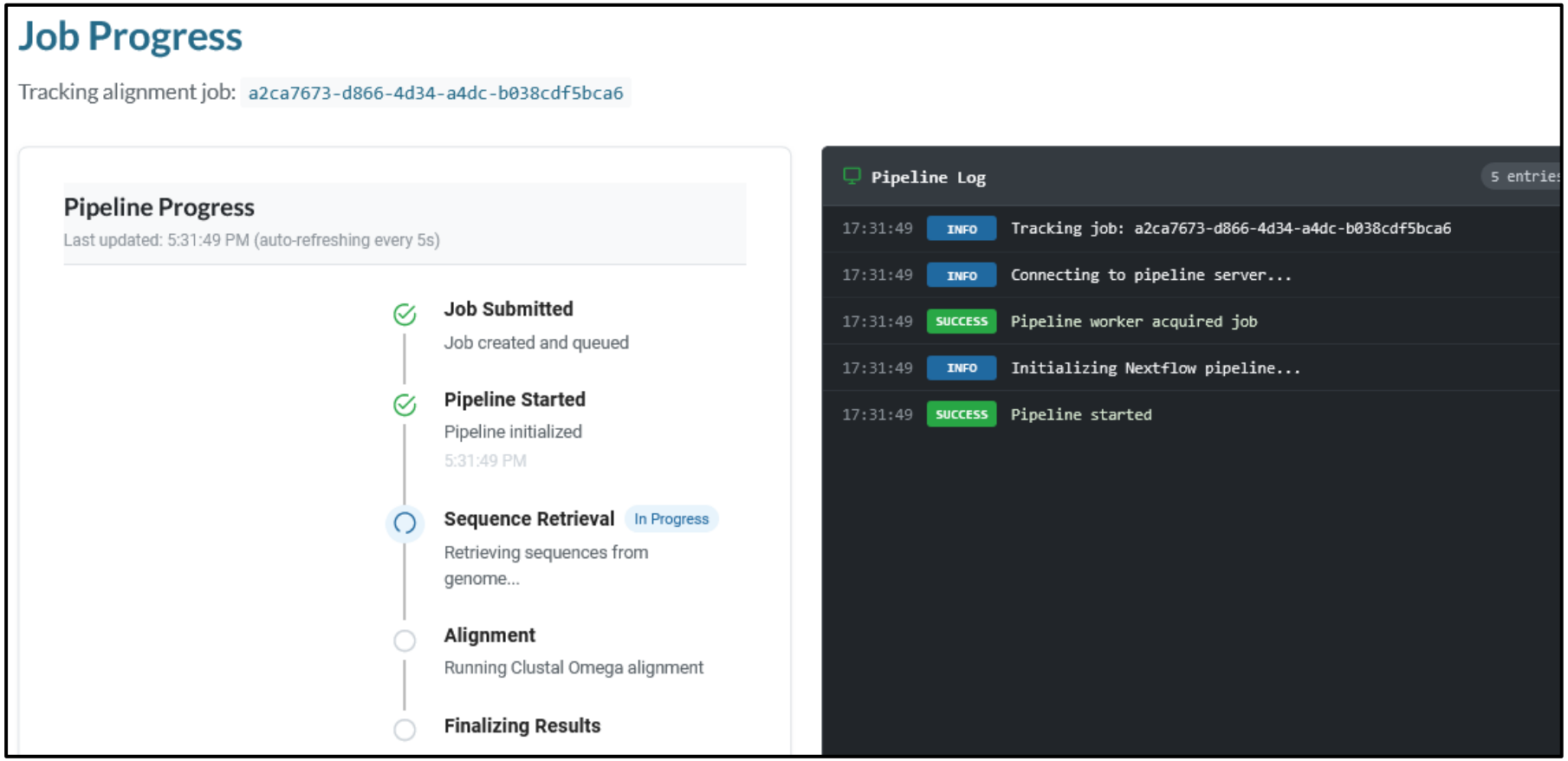
Gene (species)	**Transcript**	**Variant(s)**
Adcy3 (Rno)	NM_130779.2	NC_086024.1:g.32845550T>C
ADCY3 (Hsa)	NM_004036.5	NC_000002.12:g.24918642C>T NC_000002.12:g.24918630T>C NC_000002.12:g.24918626A>T
Adcy3 (Mmu)	ENSMUST00000127756	no variant
Ac3 (Dme)	FB:FBtr0085897	no variant

Or use this link to access this example in PAVI:

https://alliancegenome.org/pavi/result?uuid=f5322261-d9f7-4f04-a4bf-775ac2773123

The Alliance: PAVI Alignment Progress

The progress page follows the movement of the process through these steps in real time.



The tool uses the start and stop positions of the exons and coding sequence for each protein-coding transcript to create the transcript sequence from the genome assembly sequence, then translates it into the corresponding protein. If variants have been selected, the transcript sequence is altered based on the variant position and sequence and translated into an altered protein sequence. The reference proteins and any variant proteins are then aligned.

https://rgd.mcw.edu

The Alliance: PAVI Alignment Results

Alignment details

Variant details

Adjust color scheme

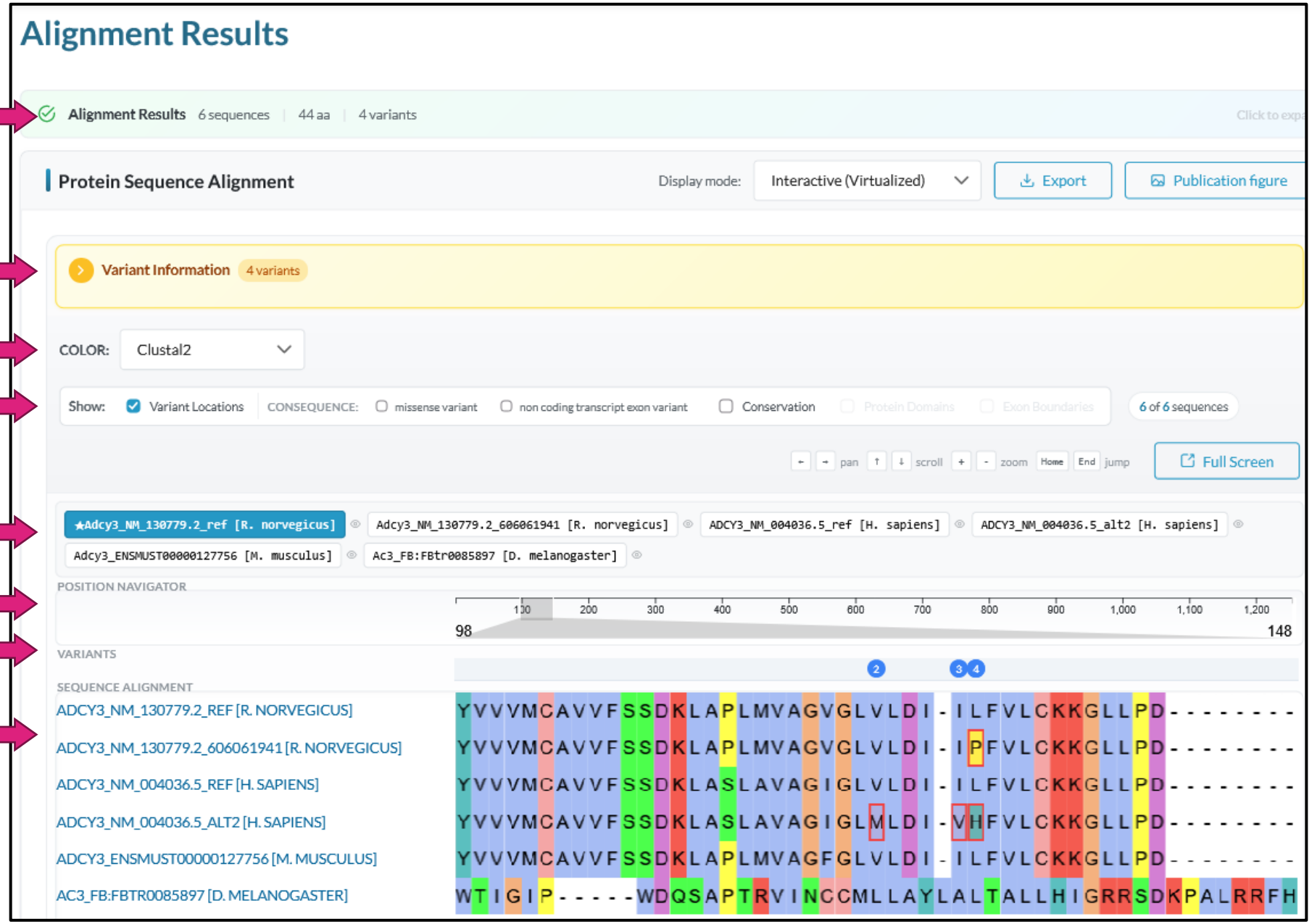
Display options

Aligned sequences

Alignment position

Variant positions

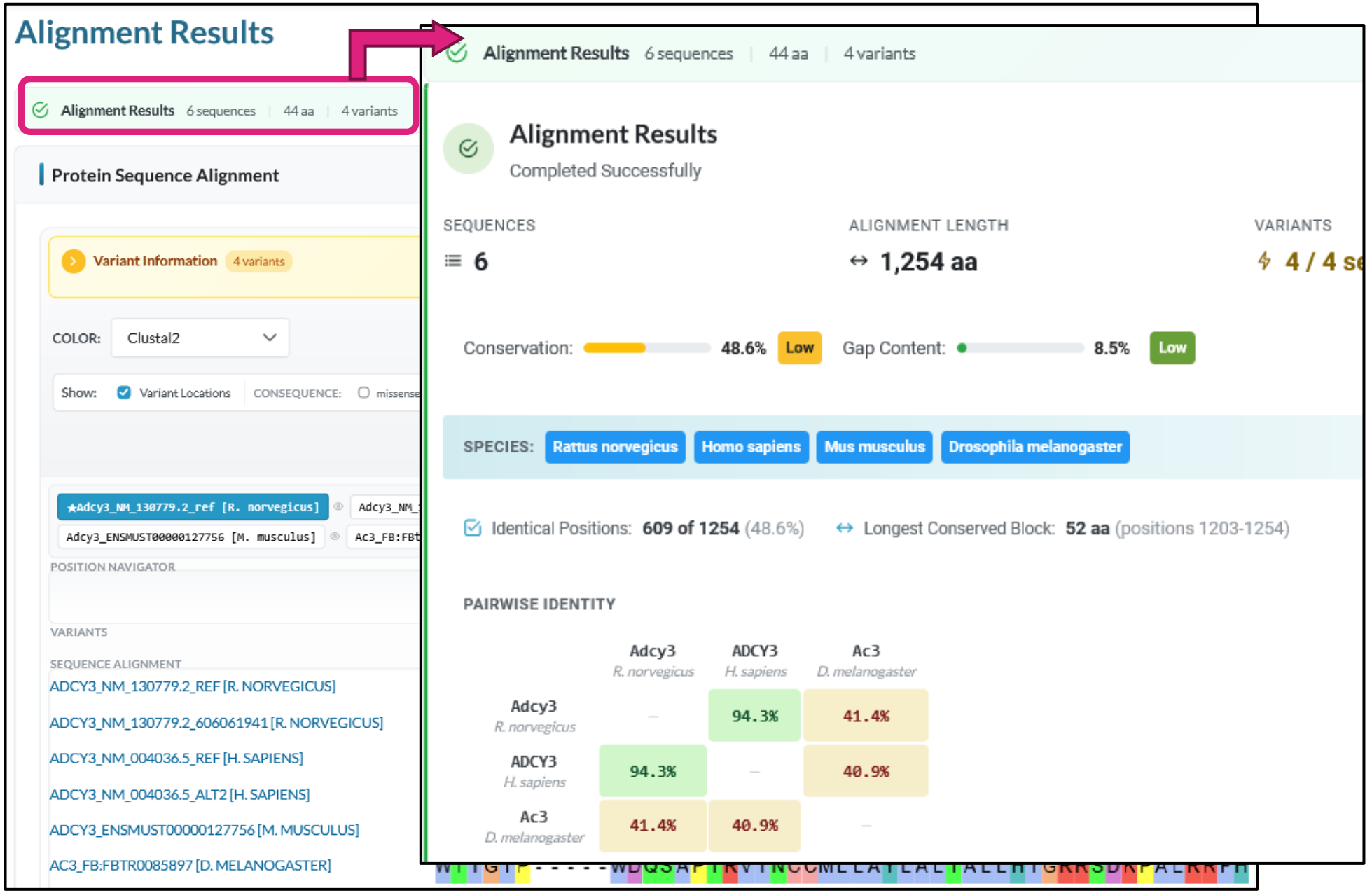
Protein sequence alignment



https://rgd.mcw.edu

The Alliance: PAVI Alignment Results

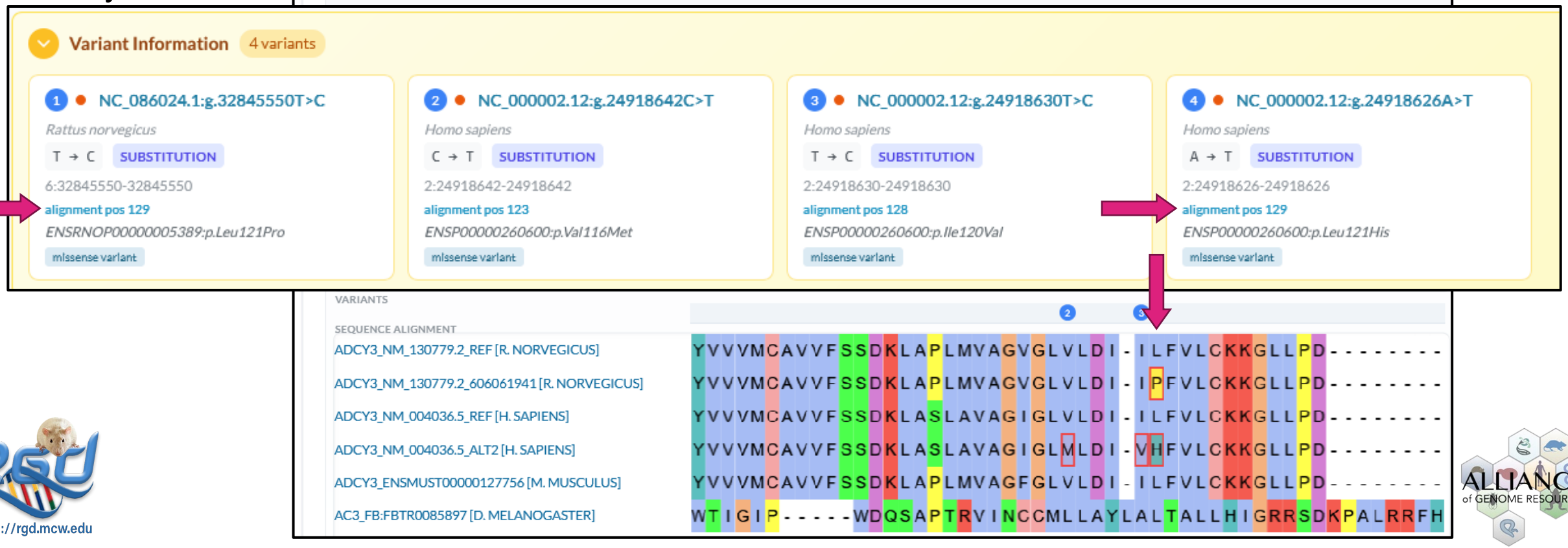
Details about the alignment are available but are not shown by default.



https://rgd.mcw.edu

The Alliance: PAVI Alignment Results

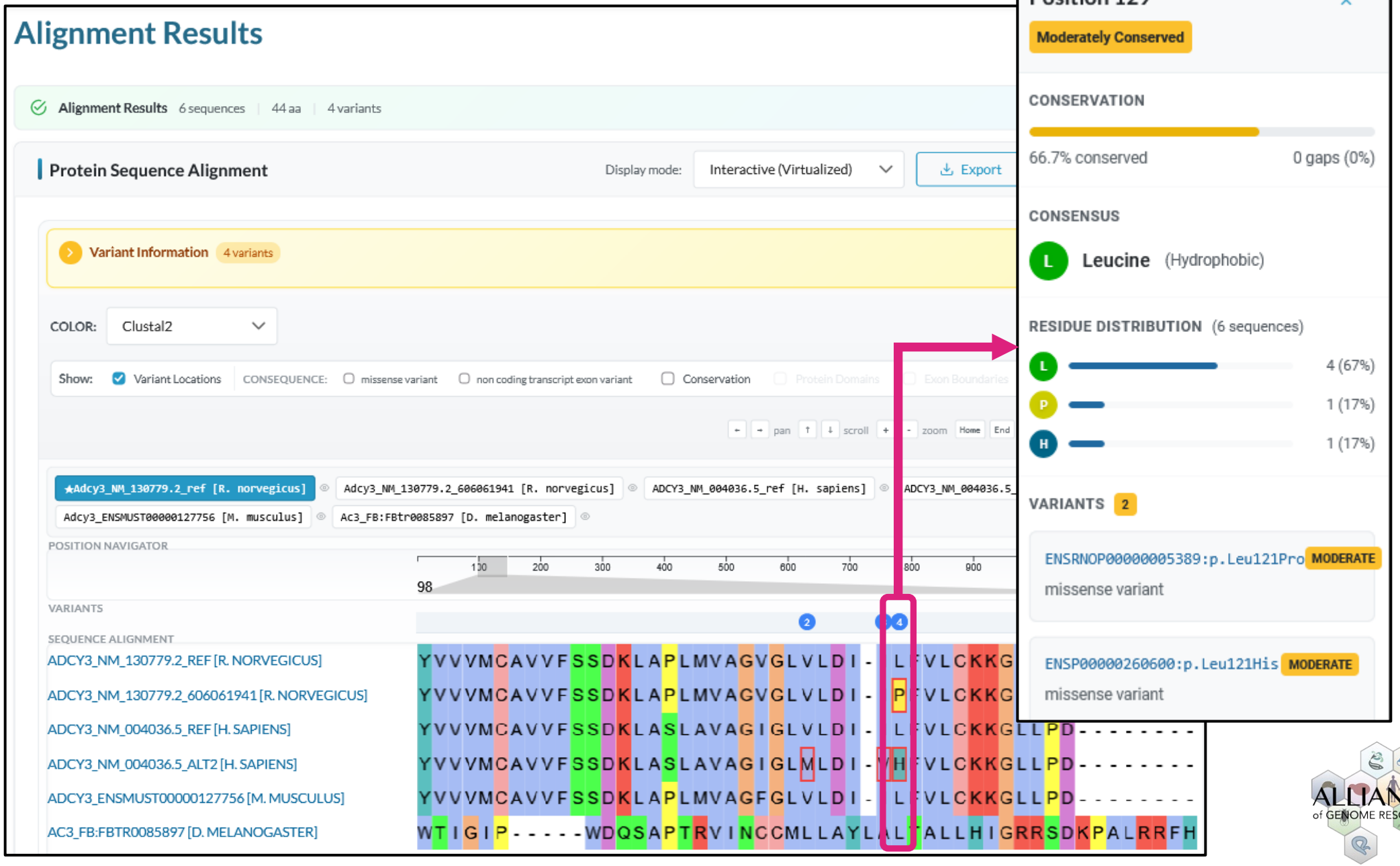
Information about the selected variants and their positions in the alignment are also available but not shown by default.



https://rgd.mcw.edu

The Alliance: PAVI Alignment Results

Click on a column in the alignment to to view a detail popup about that position.



https://rgd.mcw.edu

Thank you!

jrsmith@mcw.edu
https://rgd.mcw.edu



We acknowledge the RGD Team

(Principal Investigator: Anne Kwitek, PhD and Co-Investigator, Mindy Dwinell, PhD)

And the Alliance of Genome Resources Variants Working Group, particularly Paulo Nuin, primary developer of PAVI.

We gratefully acknowledge our funders:

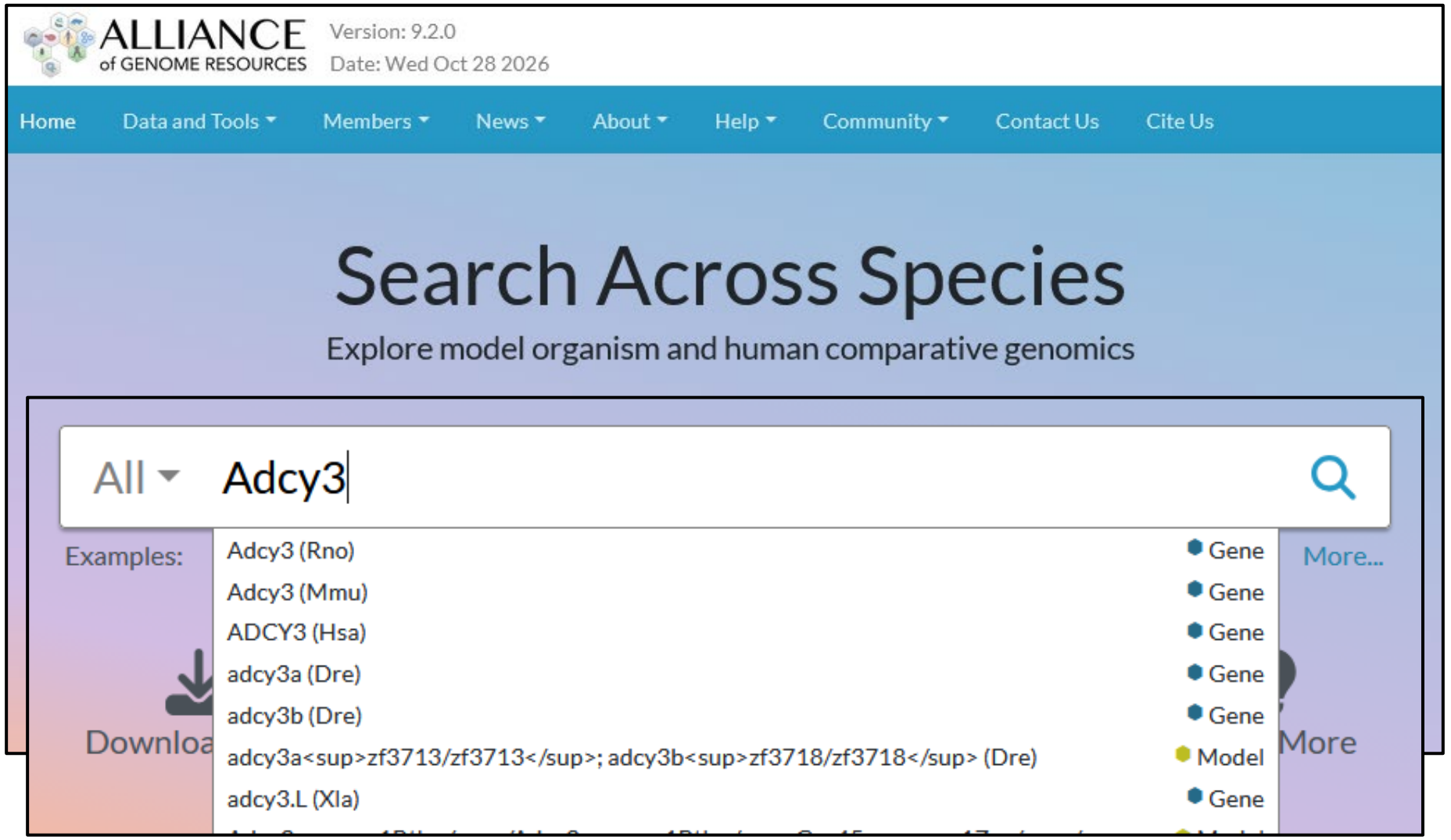
RGD is funded by the National Heart, Lung, and Blood Institute (NHLBI, R01HL064541); the Alliance of Genome Resources, and RGD as a founding member, are funded by the National Human Genome Research Institute (NHGRI, U24HG010859); the HRDP is funded by the NIH Office of the Director (R24OD024617 and R24OD039886).

We also acknowledge our collaborators and the researchers who contribute data and who faithfully use our website and data!

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The following slides give information about the Alliance gene and variant pages which may or may not be covered in the workshop, depending on time.

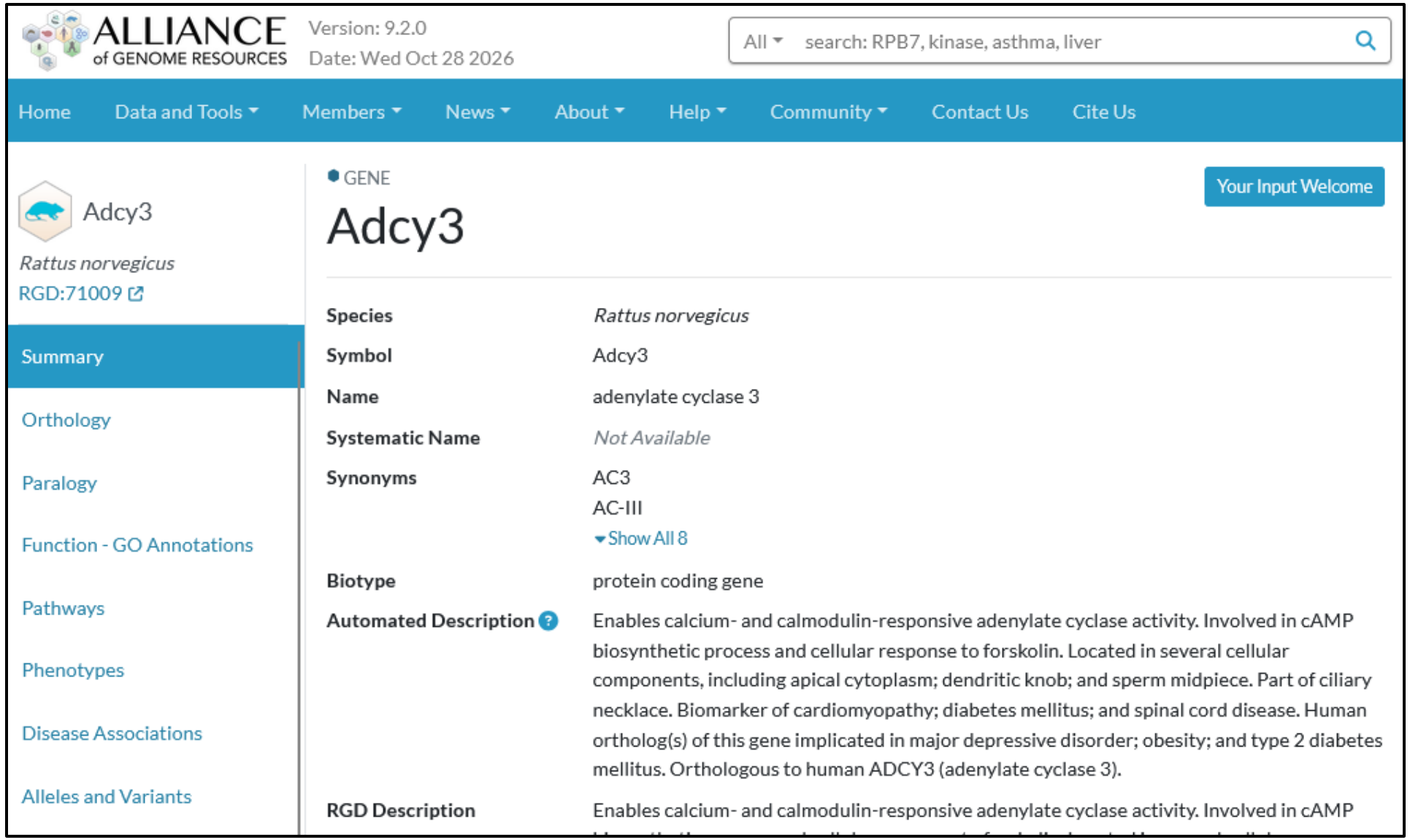
The Alliance of Genome Resources: Genes and Variants



The Alliance gene search; for this example, we will select the rat gene.

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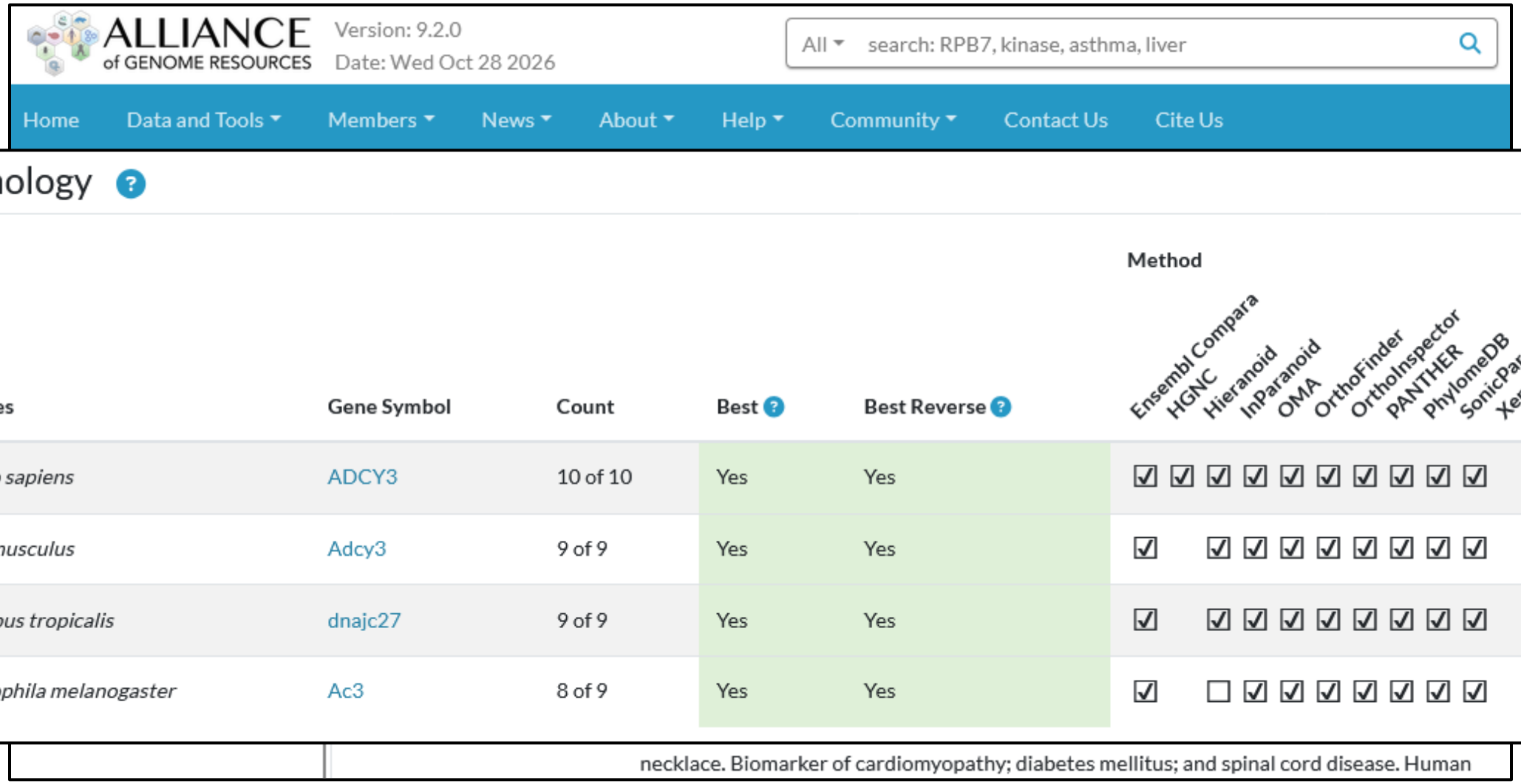
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The Alliance gene page; the summary section is shown here. Use links in the left sidebar to easily access other sections of the page.

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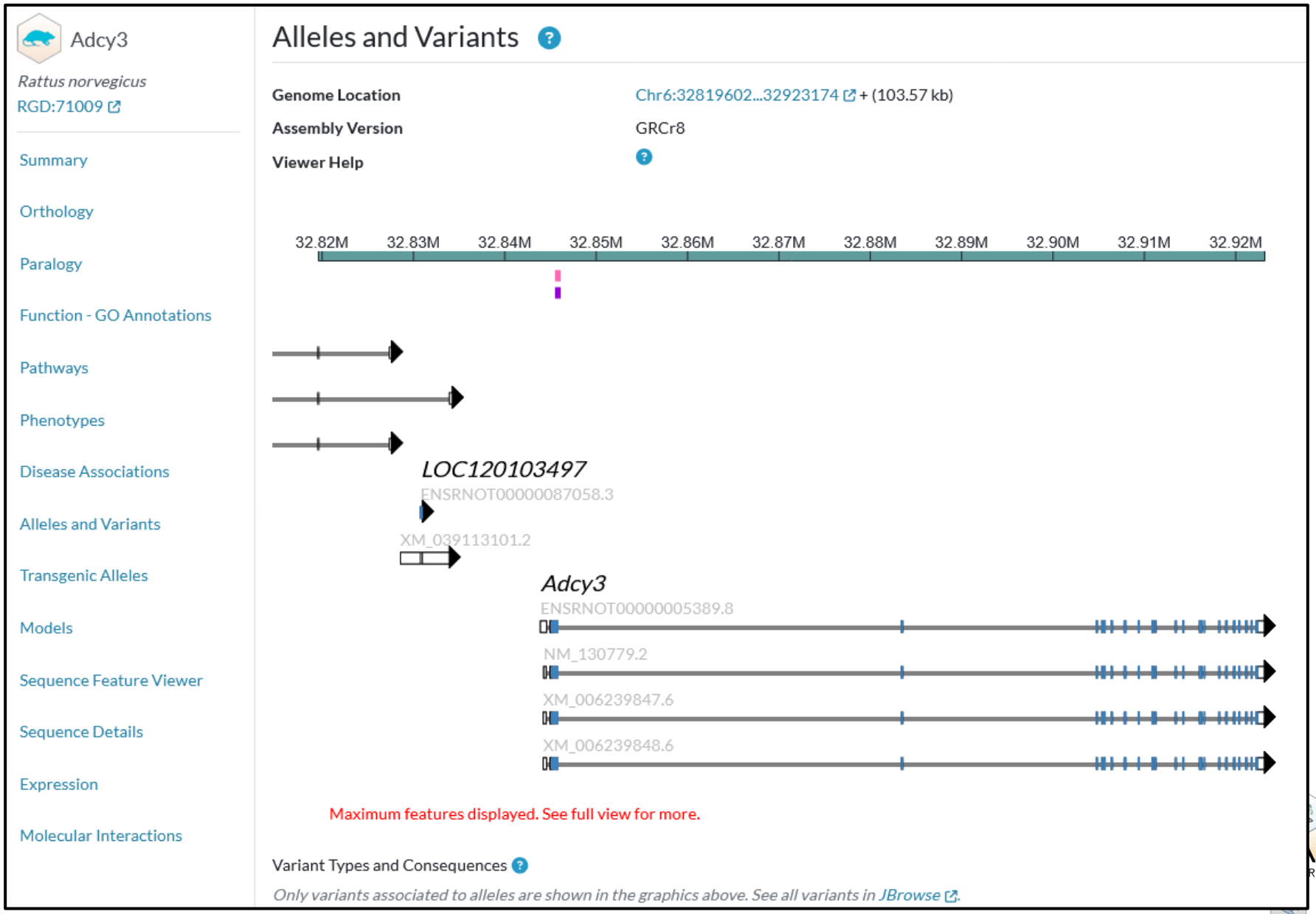


The orthology section of the rat Adcy3 gene page.

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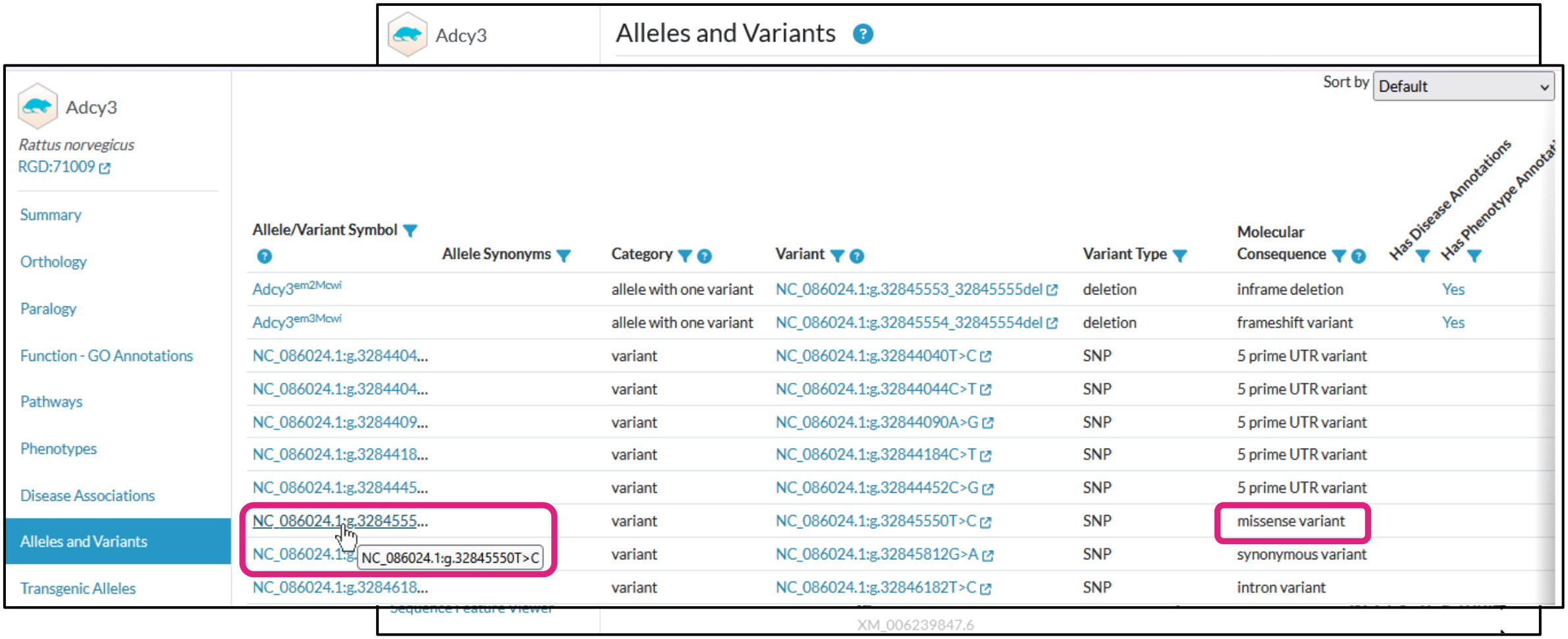
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The Alleles and Variants section of the gene page, in this case, the rat Adcy3 gene page. Shown here is the sequence viewer.



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The alleles and variants table on the rat Adcy3 gene page. The circled variant is the rat one used in the PAVI example. The name in the first column links to the allele/variant page.

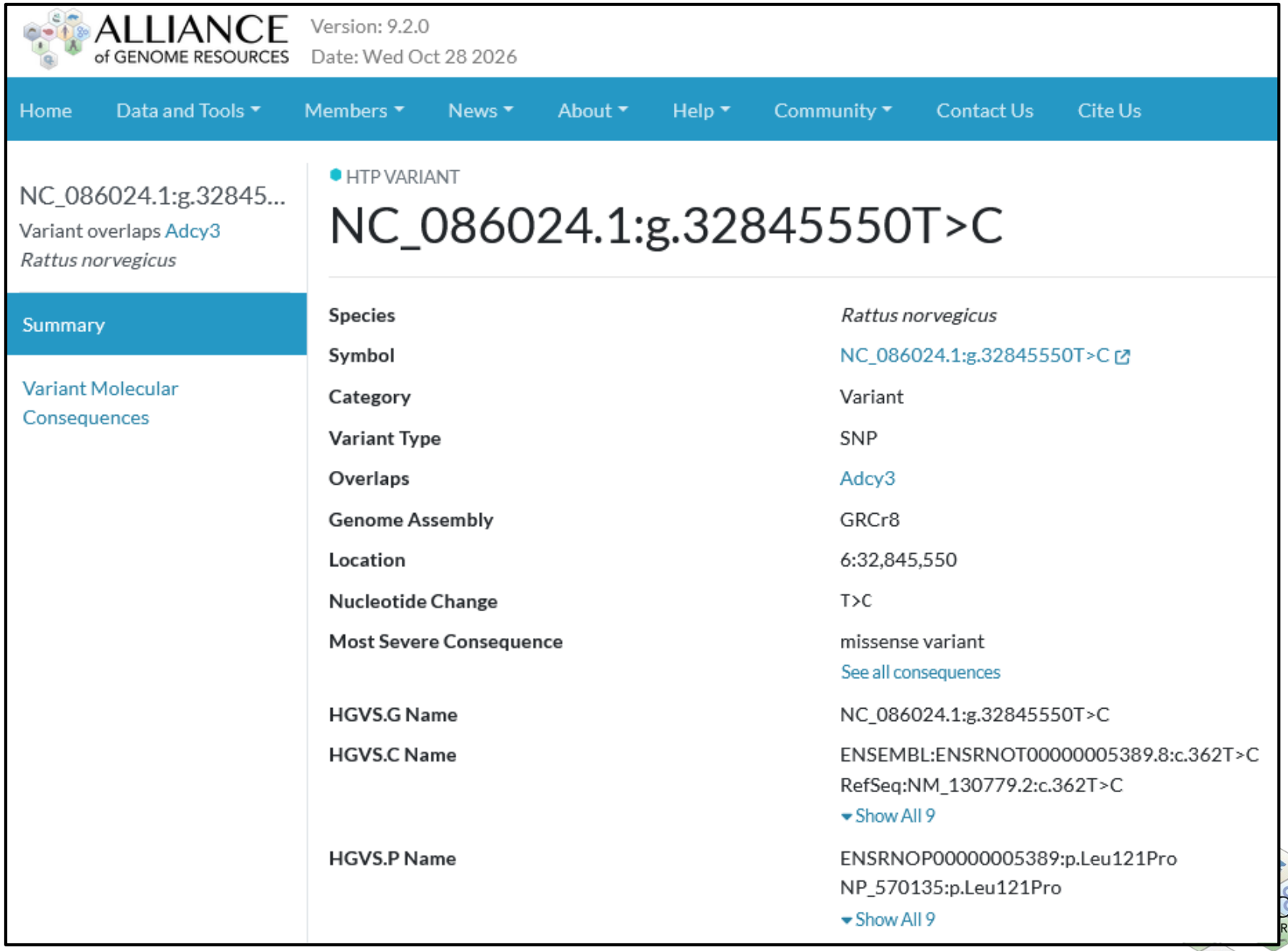
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The Allele/Variant page gives detailed information about an allele, including its variants

In the case of high throughput (HTP) variants, as shown here, details about the variant are provided and HGVS nomenclature is assigned.

HGVS nomenclature is used as the unique identifier for the variant. This variant, NC_086024.1:g.32845550T>C, is the one used in the PAVI example.



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The Alliance of Genome Resources: Genes and Variants

The Variant Molecular Consequences section shows transcript-specific effects of the variant including SIFT and Polyphen predictions from Ensembl's Variant Effect Predictor (VEP).

Variant Molecular Consequences

Hide all details	Sequence Feature	Sequence Feature Type	Associated Gene	Location	Molecular Consequence	Codon and amino acid change
Show details	ENSEMBL:ENSRNOT00000005389.8	protein_coding	Adcy3	Exon 2/22	- missense variant	361 cTt 363 => cCt [121] L [121] => P
Hide details	RefSeq:NM_130779.2	protein_coding	Adcy3	Exon 2/22	- missense variant	361 cTt 363 => cCt [121] L [121] => P

Predicted effect of NC_086024.1:g.32845550T>C on **RefSeq:NM_130779.2**

Variant Type	SNP
HGVS Names	NP_570135:p.Leu121Pro Show All 3
Affected Sequence Feature Name	RefSeq:NM_130779.2
Affected Sequence Feature Type	protein_coding
Affected Sequence Feature Accession	RefSeq:NM_130779.2
Variant Location In Feature	Exon 2/22
Molecular Consequence	missense variant
VEP Impact	moderate
SIFT Prediction (Score)	deleterious (0)
PolyPhen Prediction (Score)	probably damaging (1)
Position of Variant in Transcript	803
Position of Variant in Coding Sequence	362
Position of Variant in Protein	121
Protein Change	

VEP Impact	moderate
SIFT Prediction (Score)	deleterious (0)
PolyPhen Prediction (Score)	probably damaging (1)

Reference				Variant
361	cTt	363	=>	cCt
[121]	L	[121]		P

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